ENGINEERING PORTFOLIO

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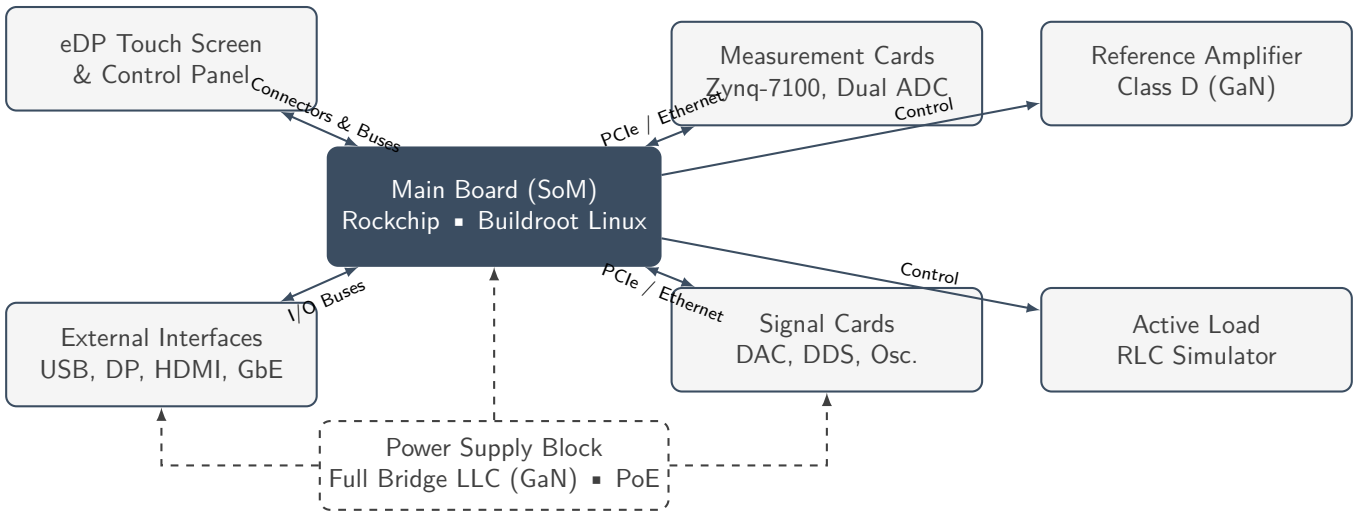
Thesis Projects

Universal Modular Measurement System

10.2025 - Ongoing

Lead Designer | Master's Thesis

Design and physical implementation of a universal, modular system for automated analysis of parameters and frequency characteristics of audio amplifiers. The platform combines digital signal processing (DSP), SoC architecture, and power electronics, enabling hardware testing under real load conditions. The system is designed to match the parameters of leading commercial equipment (e.g., Audio Precision APx555, QuantAsylum QA403).



Key Technological Solutions and Architecture:

- Central Unit (SoM): Main board featuring the Rockchip RK3588 processor (Buildroot Linux), integrating communication interfaces (PCIe, Gigabit Ethernet, USB, DP, HDMI, Wi-Fi/BT) and supporting an eDP touch screen.
- Measurement & DSP Cards (Zynq-7100): The architecture is based on two identical low-noise ADC paths. FPGAs provide hardware implementation of signal cross-correlation algorithms to eliminate equipment self-noise and nonlinearity errors.
- Active RLC Load and GaN Amplifier: A module directly controlled by the central unit, allowing simulation of real speaker parameters. The reference signal is provided by a custom Class D amplifier built on GaN transistors.
- Power Delivery: The environment is powered by the main AC/DC full-bridge LLC converter. Additionally, an auxiliary converter with a dedicated controller handles PoE standard peripherals.

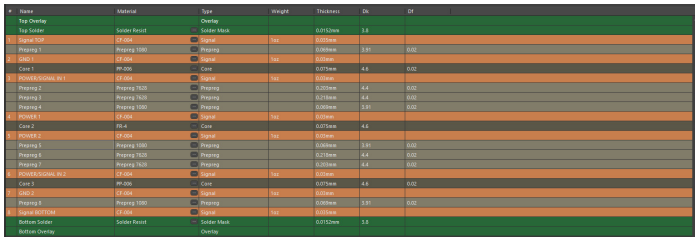
Project Status and Schedule:

- Completed stages: Full system architecture defined. Selected analog simulations performed and schematic designs finalized. Printed Circuit Boards (PCBs) designed for modules including the Ethernet TSN node and the PoE power supply.
- Current work: PCB layout routing of the computing core (Rockchip SoM) is underway. Modules such as the GaN amplifier and main LLC converter are in the conceptual phase.
- Detailed subsystem status: A comprehensive progress report on each subsystem can be found in the subsequent sections of this document.

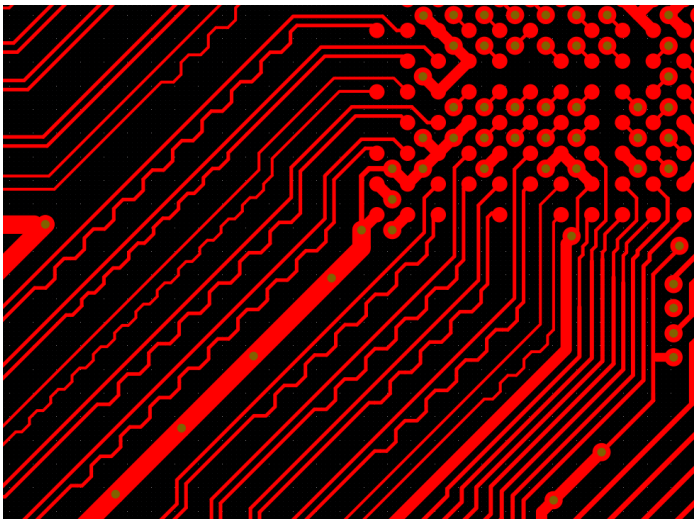
Subsystem: System on Module (SoM) – Rockchip RK3588

Architecture and Power Supply: The heart of the computing system is a custom SoM based on the high-performance Rockchip RK3588 application processor. The module integrates two LPDDR4 memory chips, QSPI memory, eMMC storage, and an advanced power delivery section. Power distribution relies on a dedicated PMIC and auxiliary DC/DC converters supplying the GPU and CPU cores. The entire board is powered by a three-phase converter stepping down the 12V input.

PCB Stackup and Impedance: The 8-layer PCB was designed and optimized for the JLCPCB manufacturing process and cost-efficiency. Using thin prepreg dielectric layers allowed for achieving target wave impedances for single-ended signals and differential pairs. This enabled relatively narrow trace widths, which proved crucial for successful fan-out of all high-speed interfaces from beneath the dense BGA footprint.

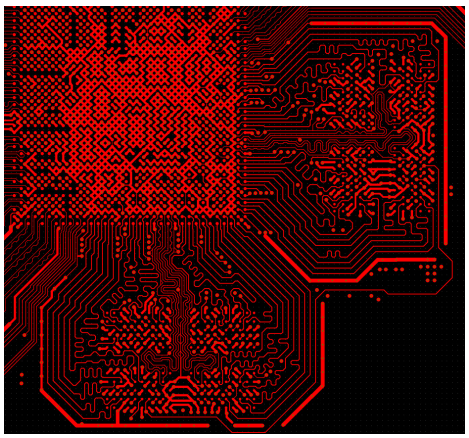


8-layer stackup (Altium Designer)

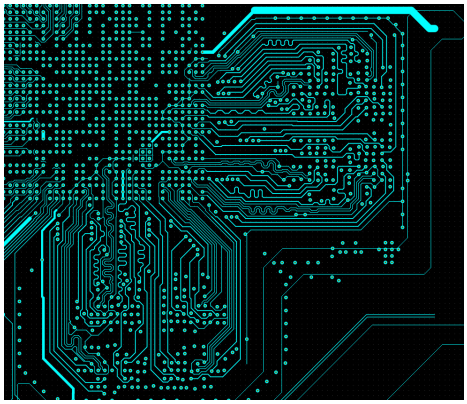


Length tuning (meandering) for the PCIe bus

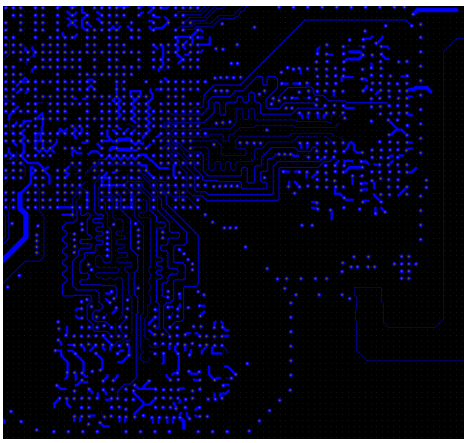
High-Speed Routing and LPDDR4 Memory: Design Rules definition and full routing of the LPDDR4 memory bus have been completed. Using the xSignals tool in Altium Designer, traces were routed taking into account the physical die/package lengths inside the processor and memory chips. Precise length tuning was achieved on respective signal layers. High-speed interfaces (PCIe, RGMII, USB 3.0, DSI, HDMI) were also routed out, minimizing phase shifts in differential pairs.



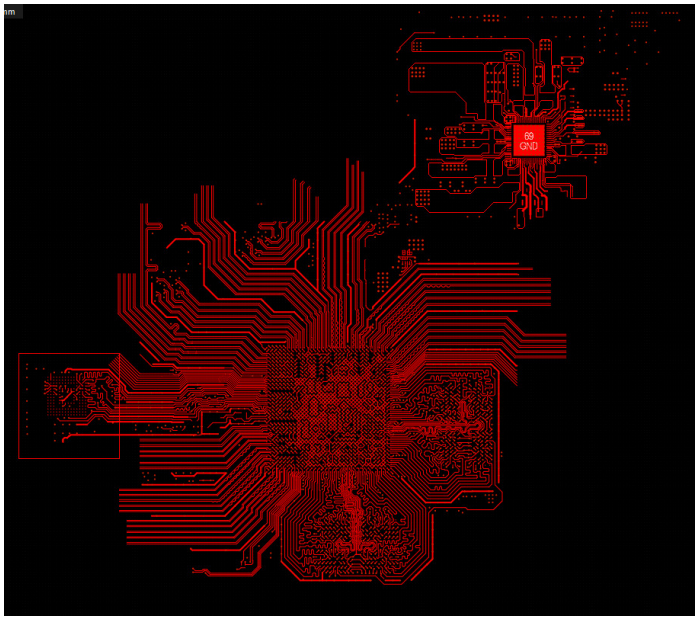
LPDDR4 - Top Layer



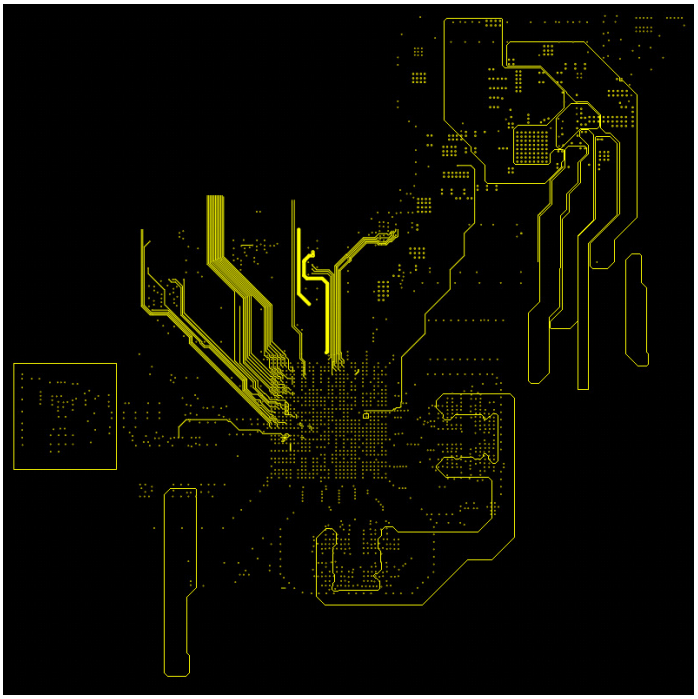
LPDDR4 - L6 Layer



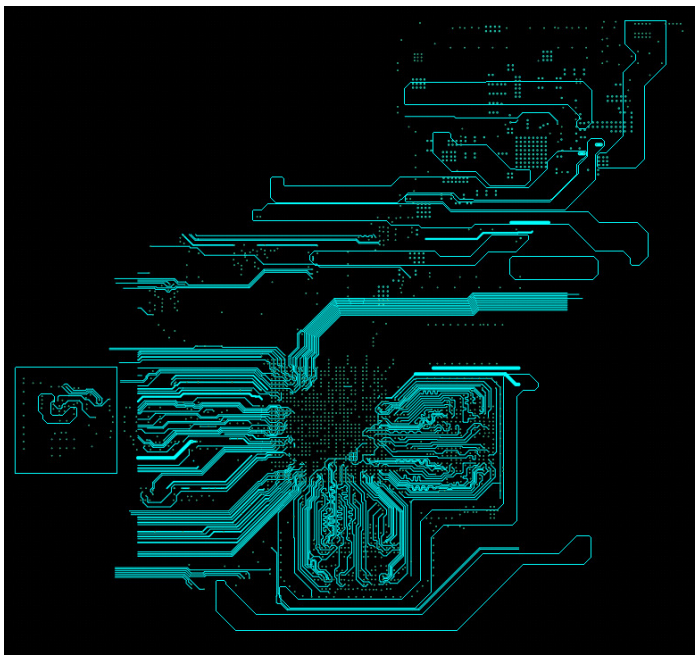
LPDDR4 - Bottom Layer



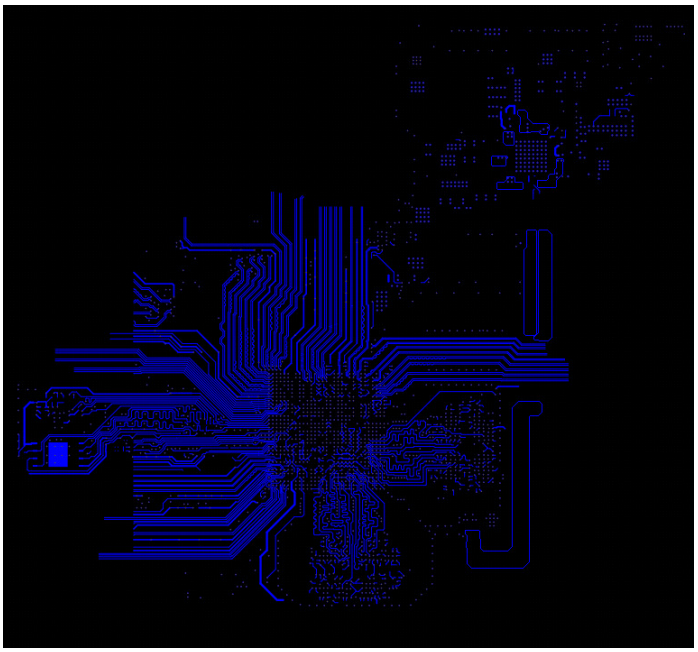
General View - Top



General View - L3

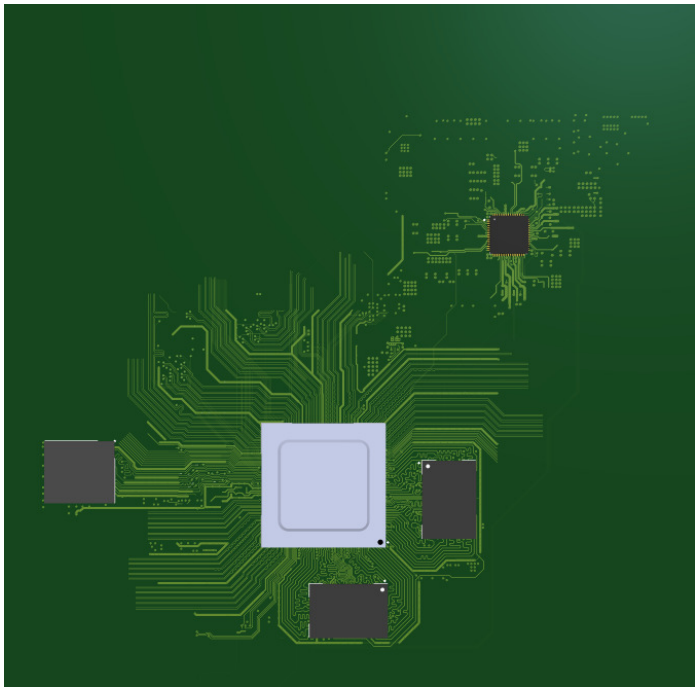


General View - L6

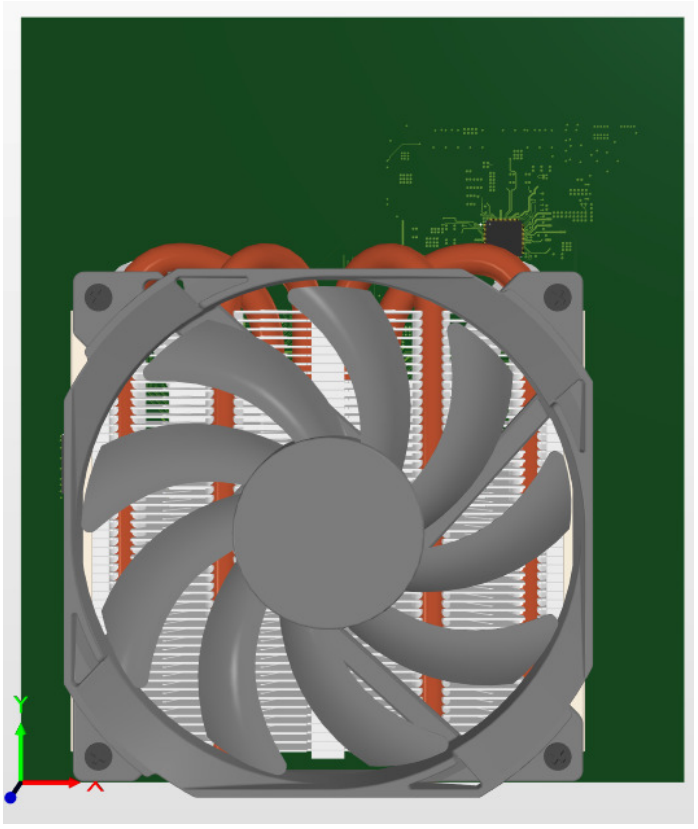


General View - Bottom

Mechanical Integration and Thermal Management (3D Model): Due to the high computational power of the RK3588 processor, the PCB design was strictly integrated with the mechanical cooling model. The placement of key ICs on the top layer (SoC, LPDDR4, PMIC) and the bottom layer organization were optimized for heat flow and assembly. To prevent mechanical stress and laminate bending, a dedicated backplate is provided on the bottom of the board. It ensures stable mounting of the massive active cooling system (heatsink with copper heat pipes and a fan) located on the main computing unit.

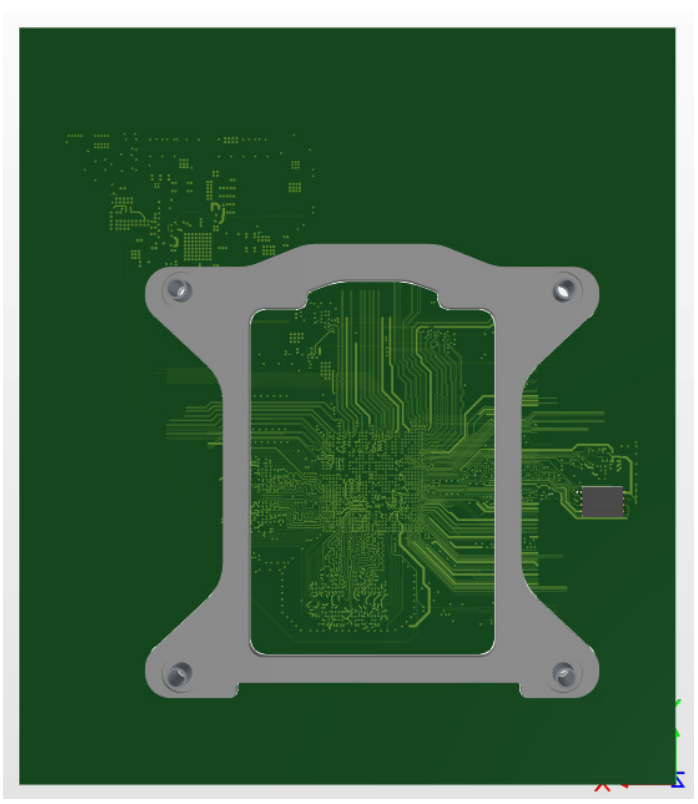


3D Model (Top) - IC Placement



Integrated active cooling

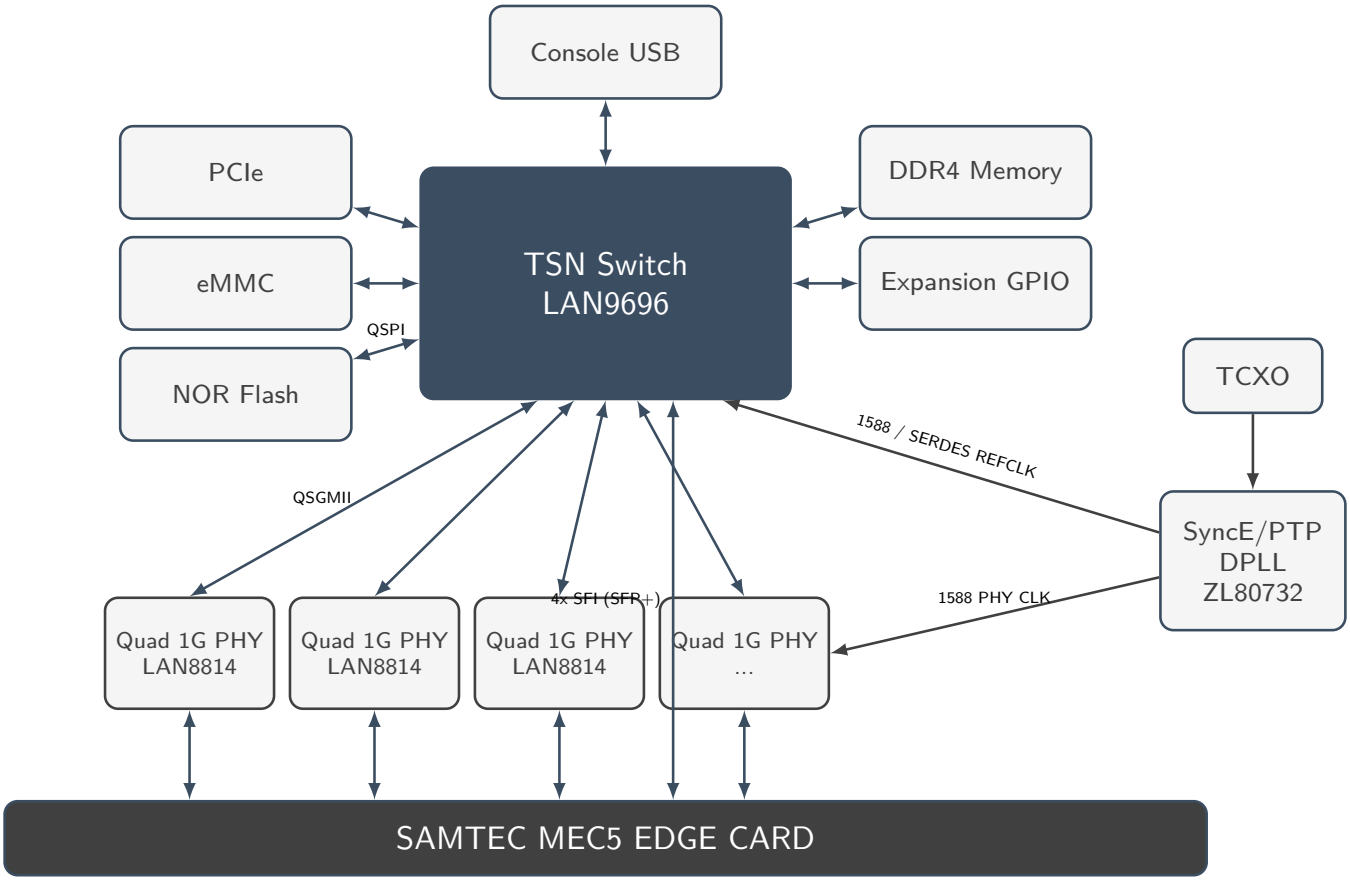
3D Model (Bottom) - Routing



Stiffening Backplate

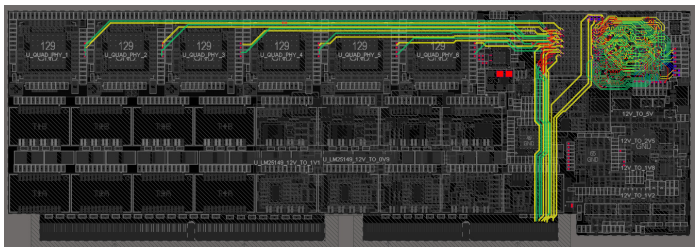
Subsystem: Ethernet TSN Expansion Card (IEEE 1588v2)

Architecture and Real-Time Communication: The expansion card is designed to provide deterministic network communication between external switches, measurement cards, and the main SoM. The core of the circuit is an advanced TSN (Time-Sensitive Networking) switch, LAN9696. To guarantee absolute real-time delivery of measurement data, hardware support for the PTP protocol (IEEE 1588v2) will be implemented. Precise time synchronization is managed by a dedicated SyncE/PTP DPLL (ZL80732) working with a high-quality oscillator. The physical layer (PHY) is handled by six quad LAN8814 chips communicating with the switch via QSGMII buses, complemented by four SFP+ (SFI) interfaces. Communication with the system backplane utilizes a SAMTEC MEC5 edge connector.

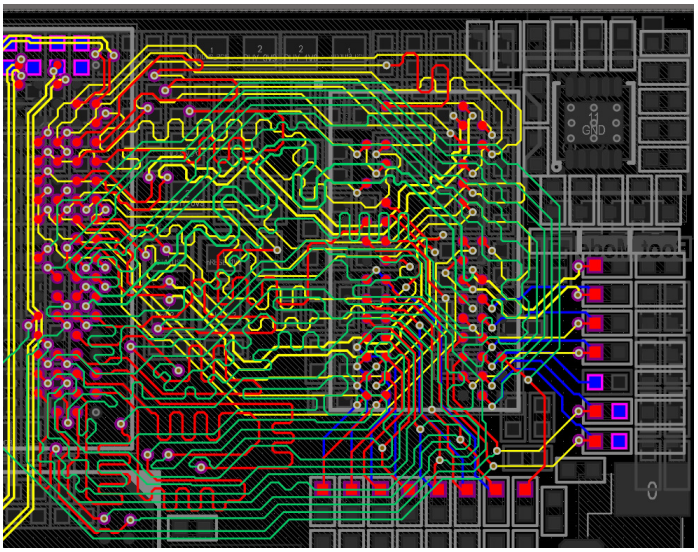


PCB Design and Signal Integrity: I am designing the expansion card PCB as a 6-layer structure. Given the significantly larger available footprint and mounting space compared to the SoM module, I opted out of the complex 8-layer stackup. The choice of an optimized 6-layer profile significantly lowers production costs while strictly meeting Signal Integrity (SI) and Power Integrity (PI) requirements. The complex power delivery block is designed to generate stable voltages for the entire circuit, including currents up to 12A for the main core.

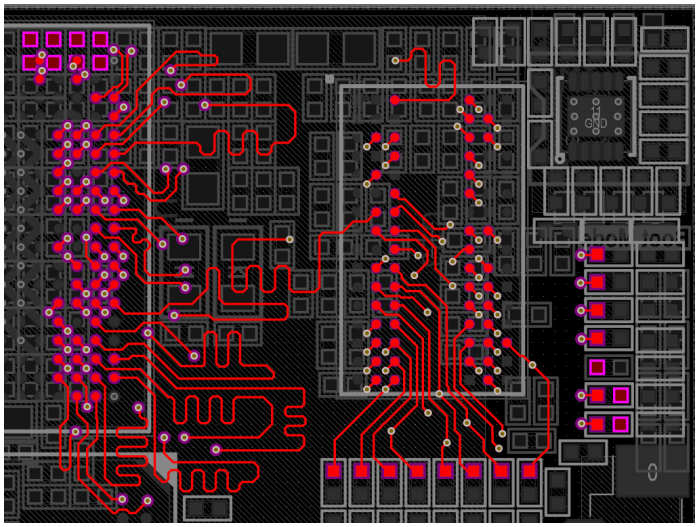
Current Work Status (Layout): I have completed the component placement and schematic designs. BGA fan-outs, including differential pairs for QSGMII and PCIe 3.0 buses, have been successfully executed. I finished the full routing of the high-speed DDR4 memory, using precise length tuning on specific signal layers to guarantee proper timing margins for data transfer. Full PCB routing is currently in progress.



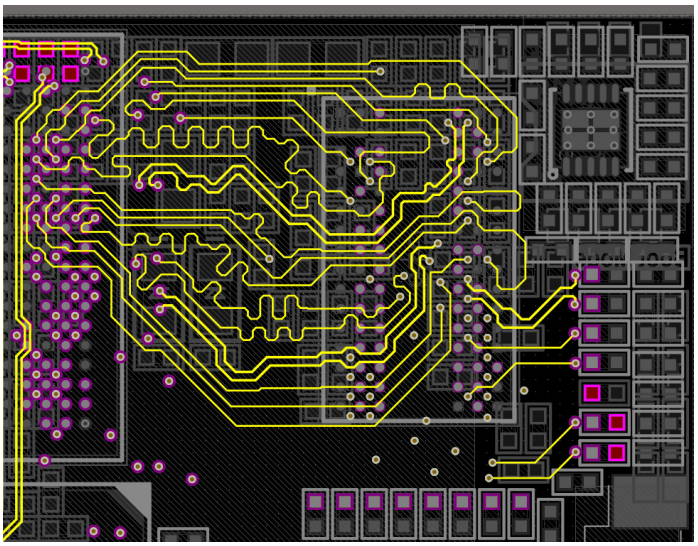
General view of signal layers (QSGMII and PCIe buses)



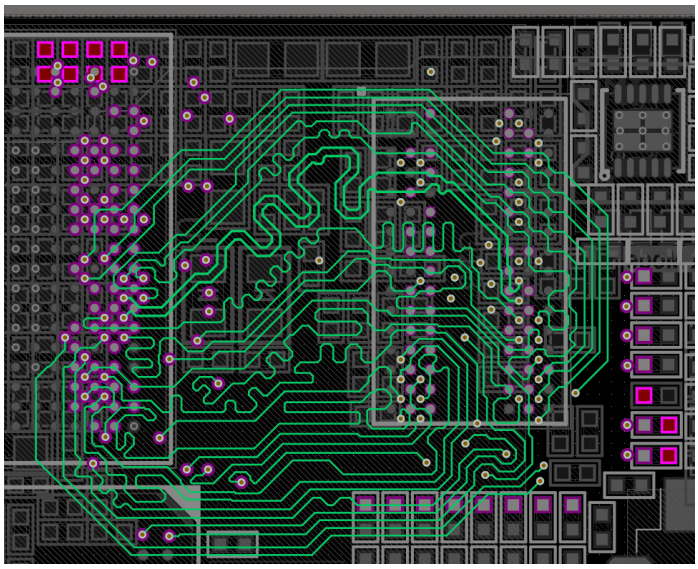
DDR Memory Routing - All layers



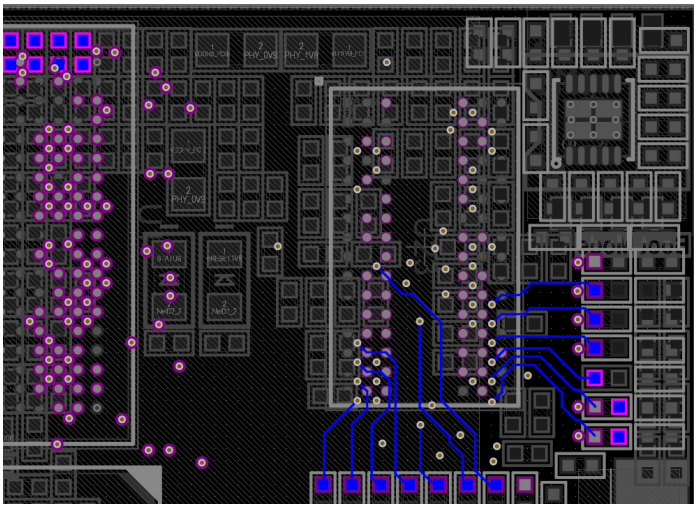
DDR - Layer 2 (Red)



DDR - Layer 3 (Yellow)

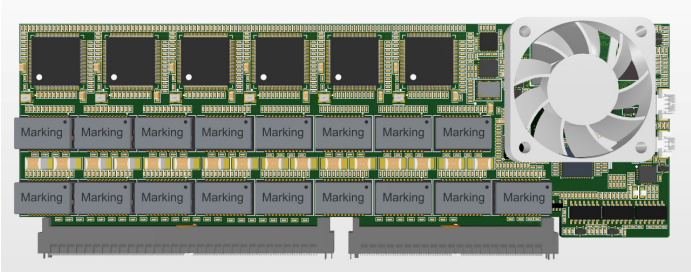


DDR - Layer 4 (Green)

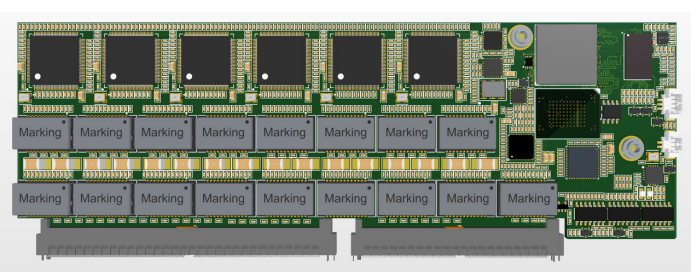


DDR - Layer 5 (Blue)

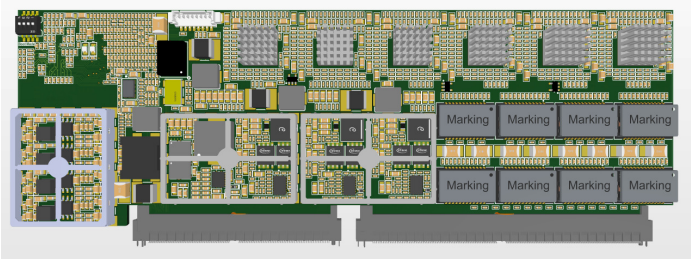
Spatial and Mechanical Integration (3D Model): The 3D model below illustrates component packaging and mechanical interface placement. The card is equipped with two massive SAMTEC edge connectors to ensure a stable connection with the backplane. The main switch IC with memories, a series of six quad PHY chips (LAN8814), and corresponding shielded signal transformers are placed on the Top layer. The extensive power supply section, covered with dedicated magnetic shields, as well as heatsinks dissipating heat from the PHY chips, are located on the Bottom layer. Such spatial arrangement guarantees extremely short signal traces between key network nodes and optimal thermal management.



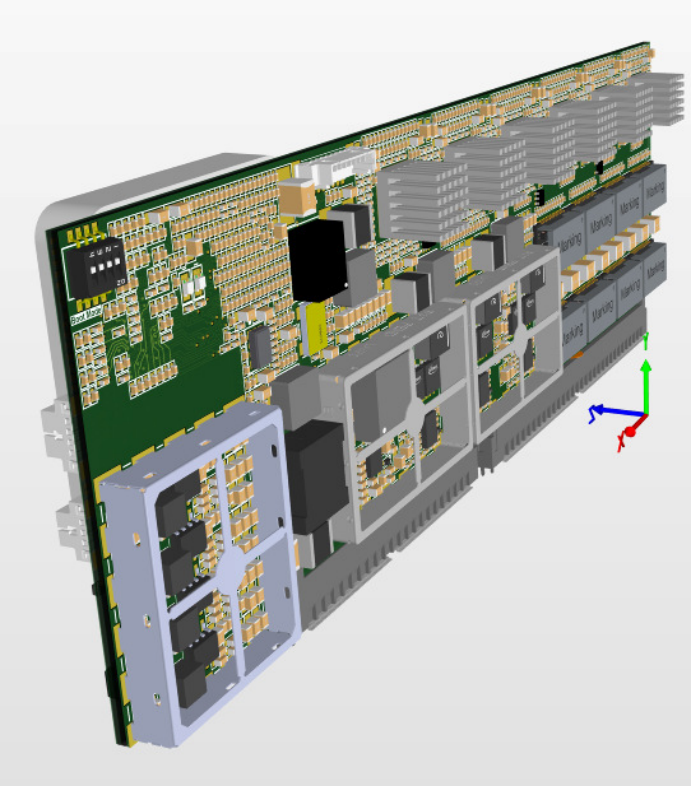
3D Model (Top) - General view with active cooling



3D Model (Top) - Perspective view of edge connectors



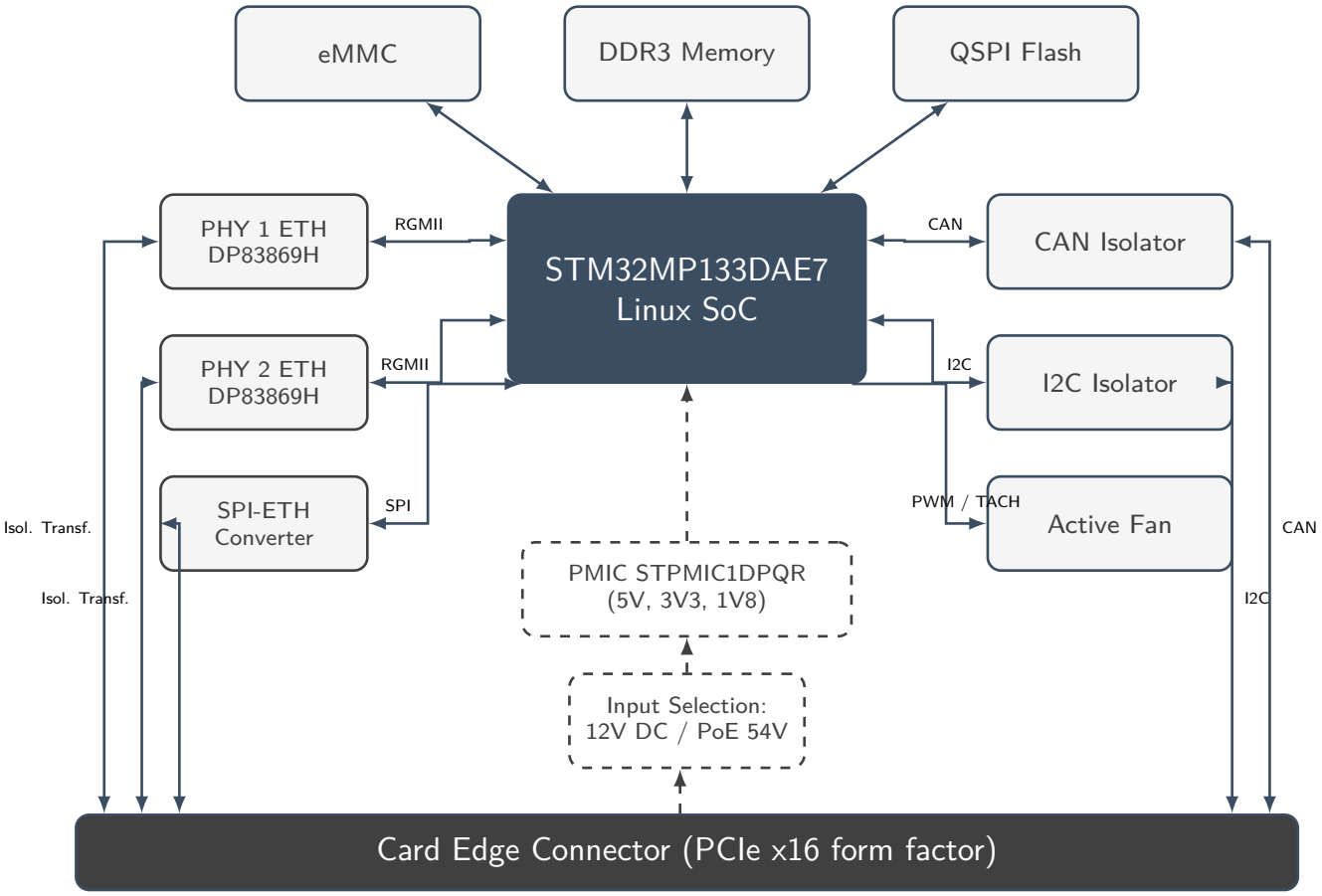
3D Model (Bottom) - PHY chips section (LAN8814)



3D Model (Bottom) - Profile and magnetic shields

Subsystem: Switch and PoE Power Supply Management Card

Architecture and Power Supply: This expansion card is designed for master management of the PoE power supply and communication with an external switch. The primary processing unit is the STM32MP133DAE7 microprocessor (industrial grade, running Linux OS). It serves as the central node of the system, having direct access to dedicated DDR3, QSPI, and eMMC memories. The system is powered from the backplane with 12V DC or directly from PoE (54V). Voltage conversion to logic levels (e.g., 5V, 3.3V, and 1.8V for the main SoC core) is handled by a dedicated STPMIC1DPQR PMIC.

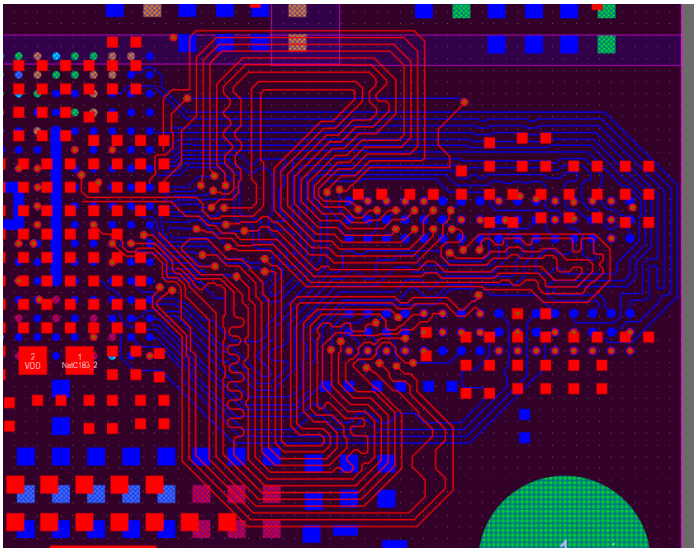


Communication and Security (Galvanic Isolation): The STM32MP1 processor has two built-in Ethernet MAC controllers that communicate via the RGMII interface with external PHY chips (DP83869HMRGZR). Network outputs are protected by full transformer isolation. To ensure reliability in industrial environments, the card handles diagnostic and operational communication with strict domain separation: a hardware CAN controller with galvanic isolation is implemented, alongside an SPI-to-Ethernet converter serving as a fail-safe channel. Communication with the master external switch is conducted over an isolated I2C bus. The microcontroller also provides thermal oversight of the device, actively controlling the integrated fan's speed via a PWM signal. Connection with the system's backplane uses a standardized PCIe x16 edge connector format.

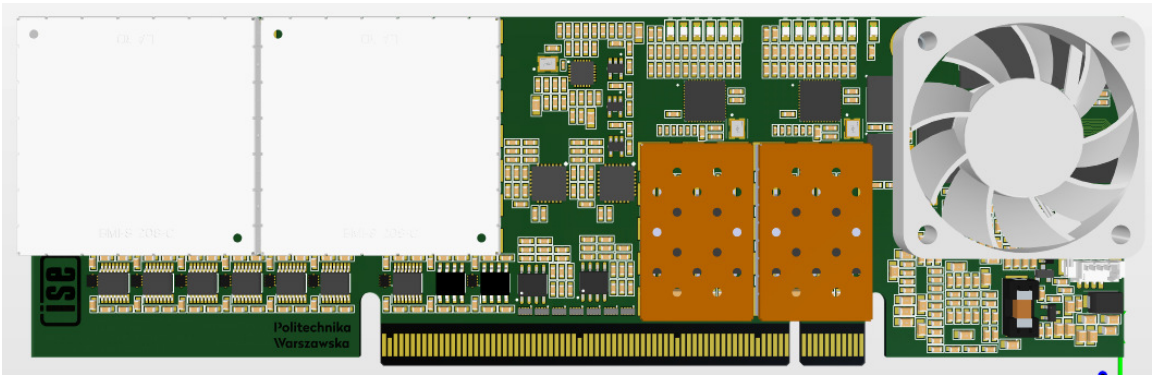
PCB Design, EMI, and Current Status: The printed circuit board was designed on a 6-layer stackup (JLC06161H-3313), reducing costs while maintaining stringent Signal Integrity (SI) requirements. To protect against RF interference and minimize electromagnetic emissions (EMI), key logic blocks and primary emission sources (including switching power supplies) were covered with dedicated mechanical RF shields. The mechanical outline and component placement have been completed, and the power supply section on the Bottom layer is fully planned. Routing of high-speed interfaces is currently in progress, including the DDR3 memory bus with precise length tuning using the xSignals tool in Altium Designer.

#	Name	Material	Type	Weight	Thickness	Dk	Df
	Top Overlay		Overlay				
	Top Solder	Solder Resist	Solder Mask		0.01016mm	3.8	
1	SIGNAL_TOP	CF-004	Signal	1oz	0.035mm		
	Prepreg_1	3313	Prepreg		0.0994mm	4.1	0.02
2	GND_1	CF-004	Signal	1/2oz	0.0152mm		
	Core_1	Core_JLCPCB	Core		0.55mm	4.6	0.02
3	POWER_SIGNAL_1		Signal	1/2oz	0.0152mm		
	Prepreg_2	2116	Dielectric		0.1088mm	4.16	
4	POWER_SIGNAL_2		Signal	1/2oz	0.0152mm		
	Core_2	Core_JLCPCB	Core		0.55mm	4.6	0.02
5	GND_2	CF-004	Signal	1/2oz	0.0152mm		
	Prepreg_3	3313	Prepreg		0.0994mm	4.1	0.02
6	SIGNAL_BOTTOM	CF-004	Signal	1oz	0.035mm		
	Bottom Solder	Solder Resist	Solder Mask		0.01016mm	3.8	
	Bottom Overlay		Overlay				

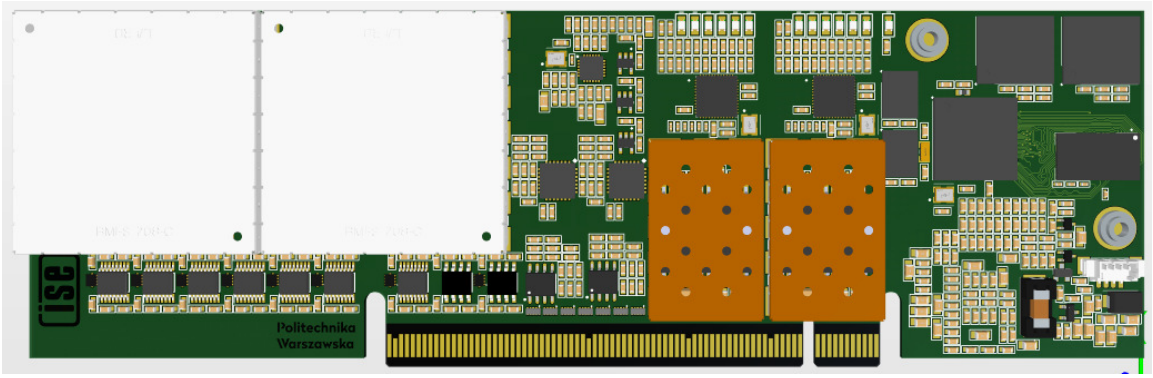
6-layer stackup (JLC06161H-3313)



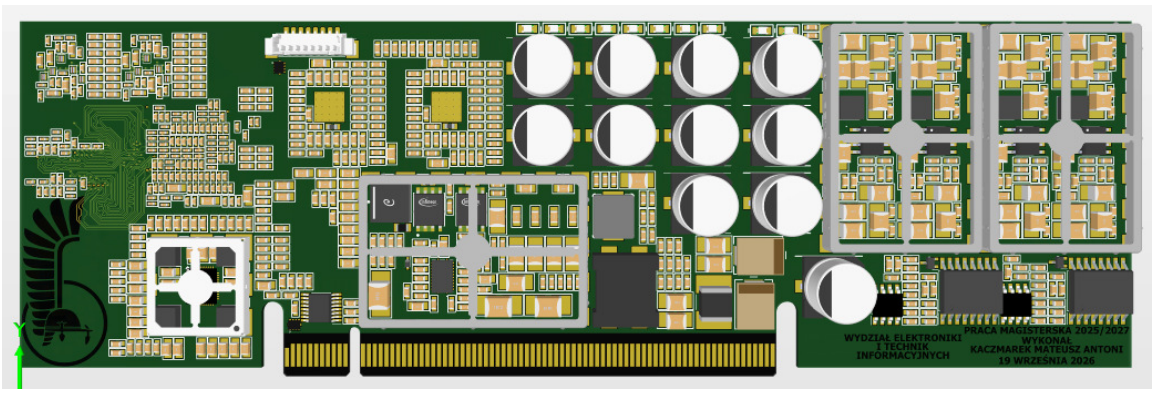
Length tuning for DDR3 memory



3D Model (Top) - Cooling and RF shields



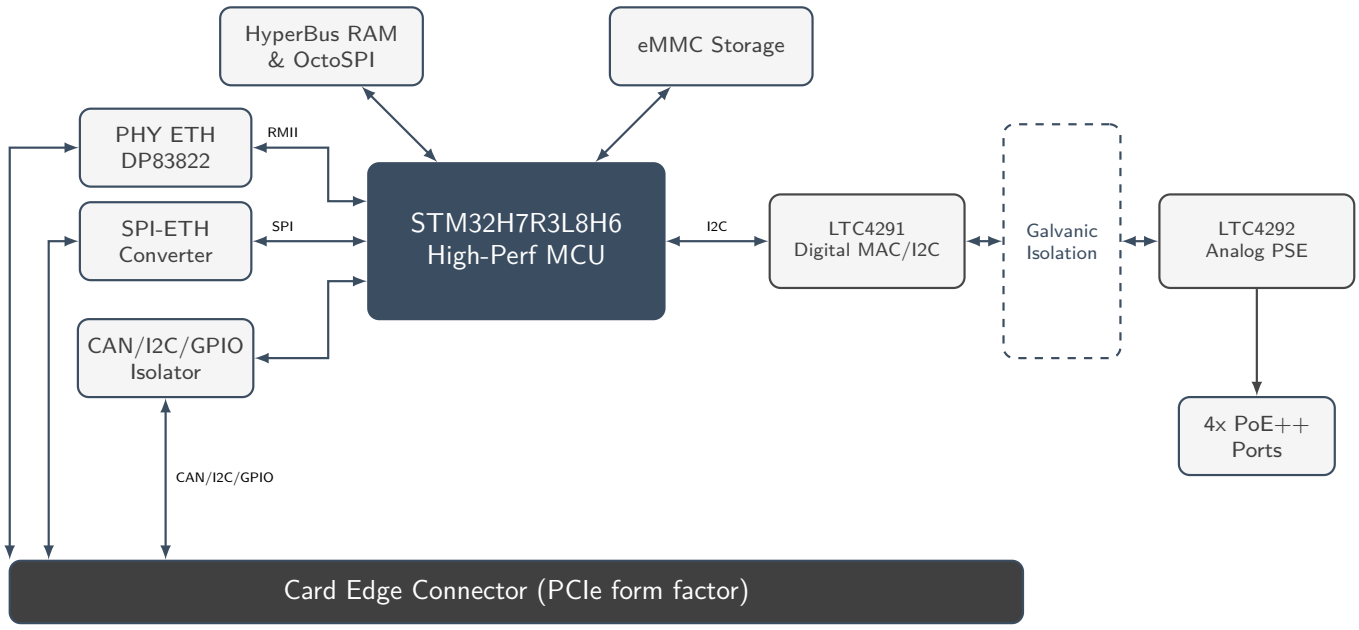
3D Model (Top) - View without the fan



3D Model (Bottom) - Power supply section

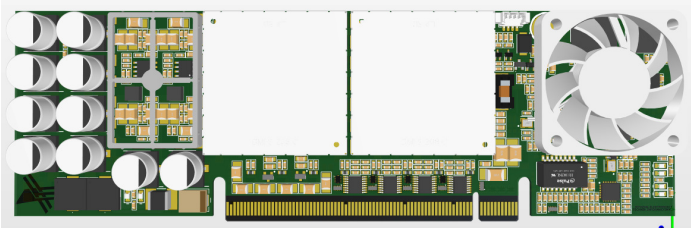
Subsystem: PoE++ Power Controller Card (IEEE 802.3bt)

Architecture and Power Management: An expansion card designed for advanced power distribution management in the PoE++ standard. Current limit control, power negotiation, and voltage monitoring are handled by a specialized PSE (Power Sourcing Equipment) chipset consisting of a pair of ICs: LTC4291IUF-1 and LTC4292IUJ. This architecture splits the controller into a digital and an analog part, allowing the implementation of a reliable galvanic isolation barrier between the control logic and the power supply output ports (54V). The circuit provides power to 4 independent high-power channels while simultaneously stepping down the PoE voltage to 5V and 3.3V logic levels using integrated DC/DC converters.

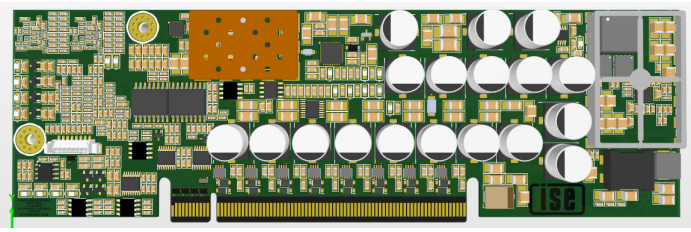


Supervision and Communication (STM32H7): The card's control system is based on the STM32H7R3L8H6 microcontroller, which utilizes external memory resources connected via ultra-fast OctoSPI and HyperBus RAM interfaces for seamless operation. An eMMC chip is used for permanent storage of logs and software. The card integrates a PHY Ethernet chip (DP83822IRHBT) complemented by an SPI-to-ETH module acting as a safety channel. Data exchange with the main board (via CAN, I2C buses, and GPIO signals) is executed with full galvanic isolation. Everything is routed to a standard PCIe edge connector.

PCB Design and Current Status: The printed circuit board relies on the same proven 6-layer stackup used in previous expansion modules, ensuring production consistency and proper parameters for high-speed signals. At this moment, schematic designs are complete, and the PCB mechanical outline has been fully defined, taking into account dedicated cooling zones and split planes (GND / ISOGND) for isolation barriers. Design rules have been set, and component placement is finalized. The board is ready for the routing stage. RF shields and a massive capacitor bank on the top and bottom layers provide stability for PoE++ standard inrush currents.



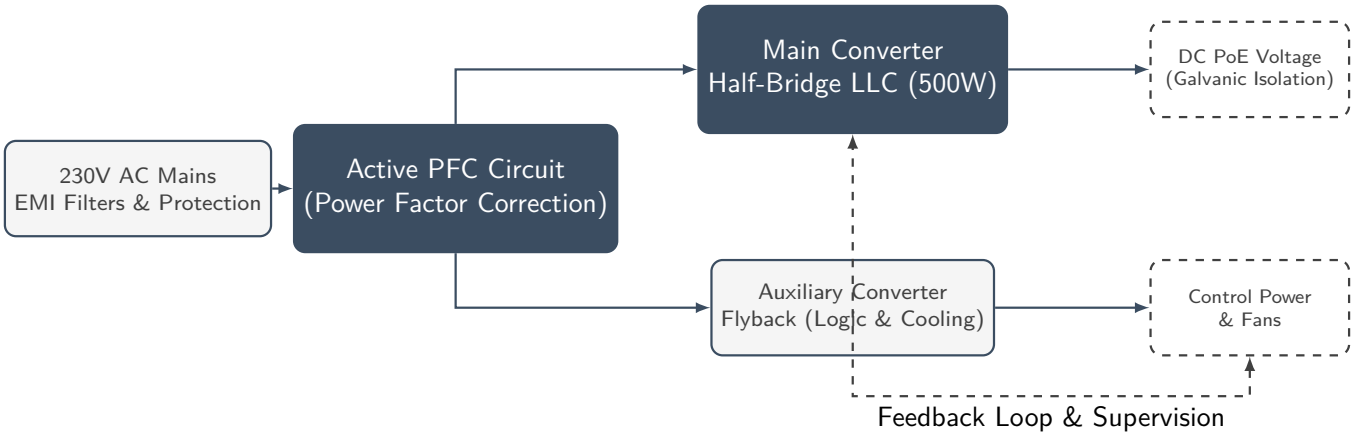
3D Model (Top) - Component placement, shields, and cooling



3D Model (Bottom) - Capacitor bank for PoE lines

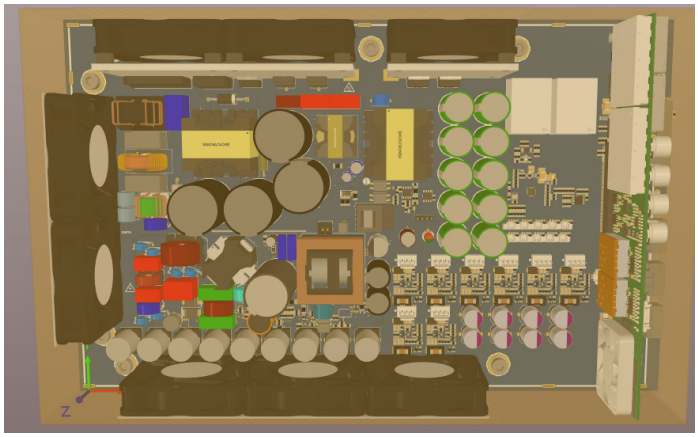
Subsystem: AC/DC PoE Power Supply (500W LLC)

Topology and Input Stage: The main PoE power source in the system was designed to convert 230V AC mains voltage into the isolated DC voltage required by the PoE standard. The input stage is equipped with a comprehensive EMI filter, an active Power Factor Correction (PFC) circuit, and a complete suite of protections: overvoltage, overcurrent (short-circuit), and electric shock protection. The PSU architecture is based on a half-bridge LLC resonant converter topology delivering 500W of continuous power. Furthermore, an auxiliary Flyback converter of lower power is integrated on the board, dedicated to powering control circuits on the primary and secondary sides, as well as the fans.

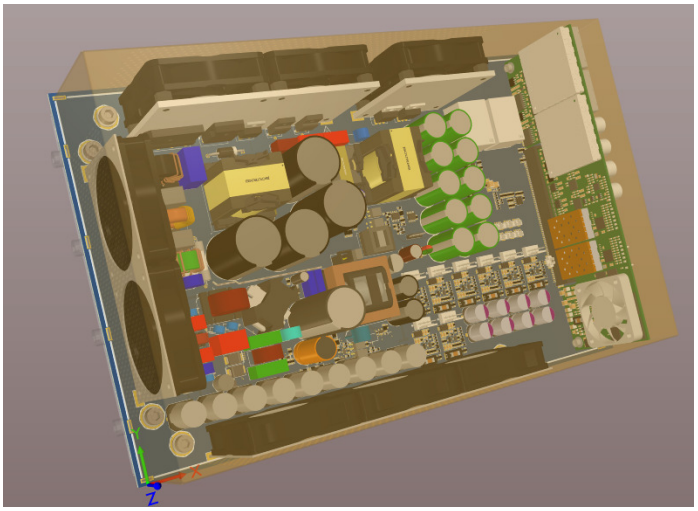


Reference Design Modification and Components: The project is based on a proven Texas Instruments reference design. I thoroughly analyzed and redesigned it to meet the voltage specifications of the PoE standard (the original design assumed a 48V output). This required recalculating and optimizing the feedback loop. Supply chain aspects were also considered – magnetic components were substituted to rely exclusively on commercially and widely available transformers and inductors, completely eliminating the problematic need to manually wind components.

Mechanical Design and Ruggedness (V-Slot): The power supply was built on a standard 4-layer PCB. Due to the target modular architecture, where external cards are plugged vertically into edge connectors, the carrier board would be prone to laminate cracking. To prevent this, the structure was integrated with a dedicated support frame made of 20x20 V-Slot aluminum extrusion profiles. The whole assembly is enclosed in a metal casing that protects against mechanical damage and acts as a Radio Frequency (RF) shield. To ensure operator safety, an additional insulating cover with cutouts for connectors is provided on the bottom of the profiles, fully preventing electric shock.

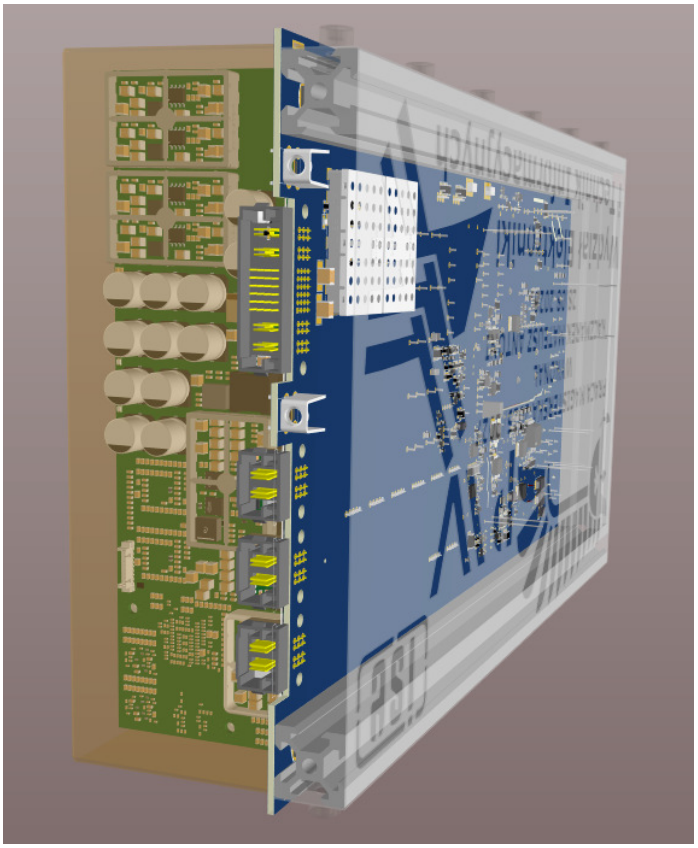


3D Model (Top) - Main filters and power components



3D Model (Top)

3D Model (Bottom) - V-Slot 20x20 structure reinforcements



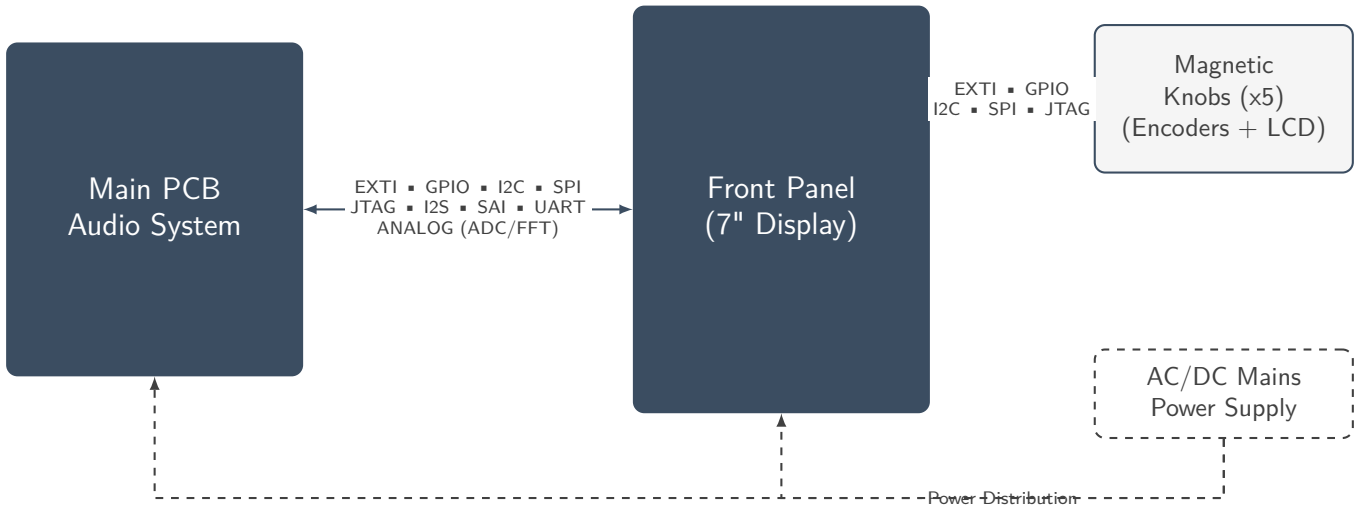
Vertically plugged edge connectors

Advanced Audio-Visual System

10.2024 - 09.2025 (Defended)

Lead Designer | Bachelor's Thesis

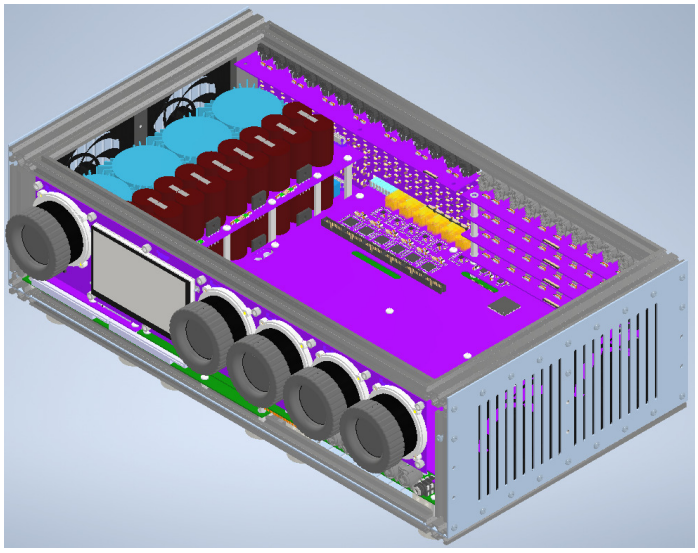
Design, construction, and physical hardware bring-up of an advanced, modular audio-visual system. The platform integrates high-end analog signal processing paths with a modern digital control system and User Interface (UI).



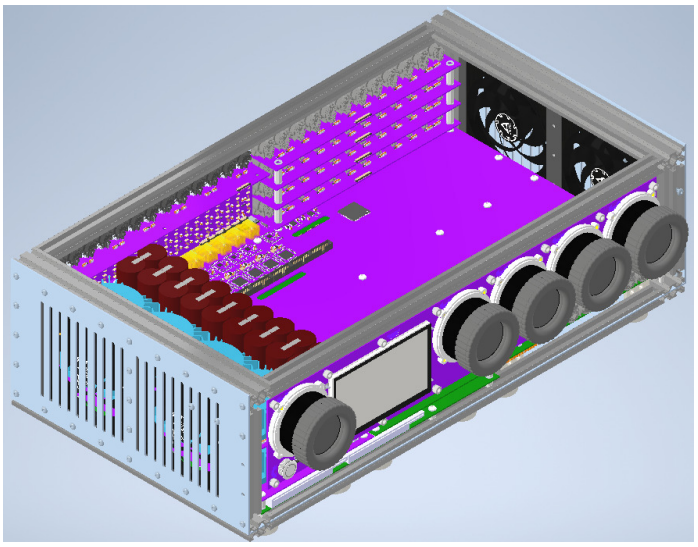
System Architecture and Communication:

- Audio Main PCB: The heart of the system responsible for signal conditioning. It connects an advanced analog section (preamplifiers, filters, equalizers) with a digital master unit managing audio routing.
- Front Panel and Magnetic Knobs: A user interface based on a 7-inch touch screen. It is complemented by magnetic knobs (functioning as contactless encoders), each with its own integrated display to present parameters (e.g., volume or selected source).
- Buses and DSP: Data exchange within the system is facilitated by a dense network of interfaces (SPI, I2C, UART). Audio signals are transmitted digitally via I2S/SAI buses to a DSP processor or analogically to ADCs for hardware calculation of the Fast Fourier Transform (FFT) and spectrum visualization.

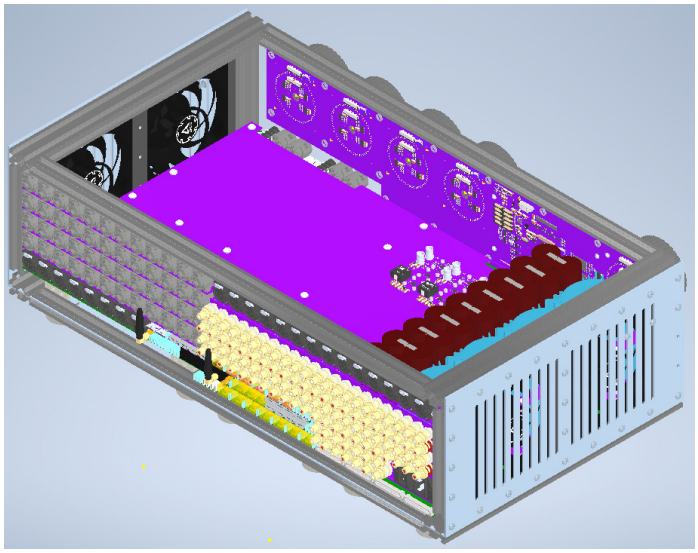
Mechanical Integration and Spatial Architecture: The construction of the device required a strict coupling of electronics design (ECAD) with mechanics (MCAD). A dedicated enclosure was designed to accommodate the extensive mainboard, massive power supply section, active cooling, and a multi-module front panel. Component placement was optimized for airflow, minimization of electromagnetic interference (separation of the power supply from sensitive audio paths), and user ergonomics.



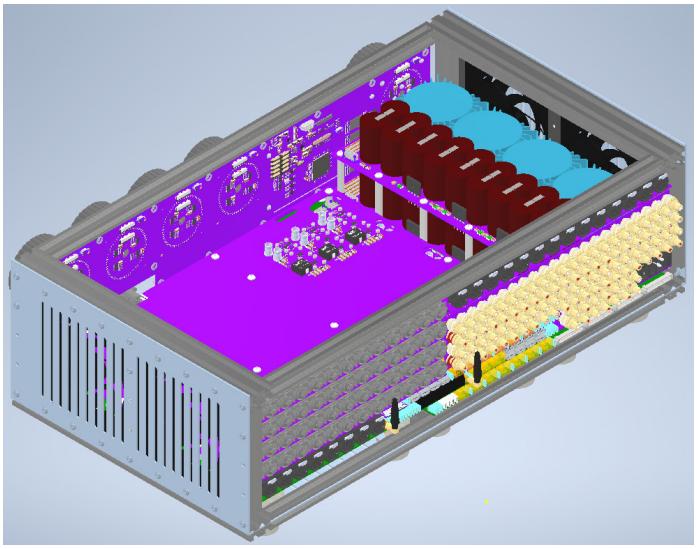
3D Model (Front-1)



3D Model (Front-2)



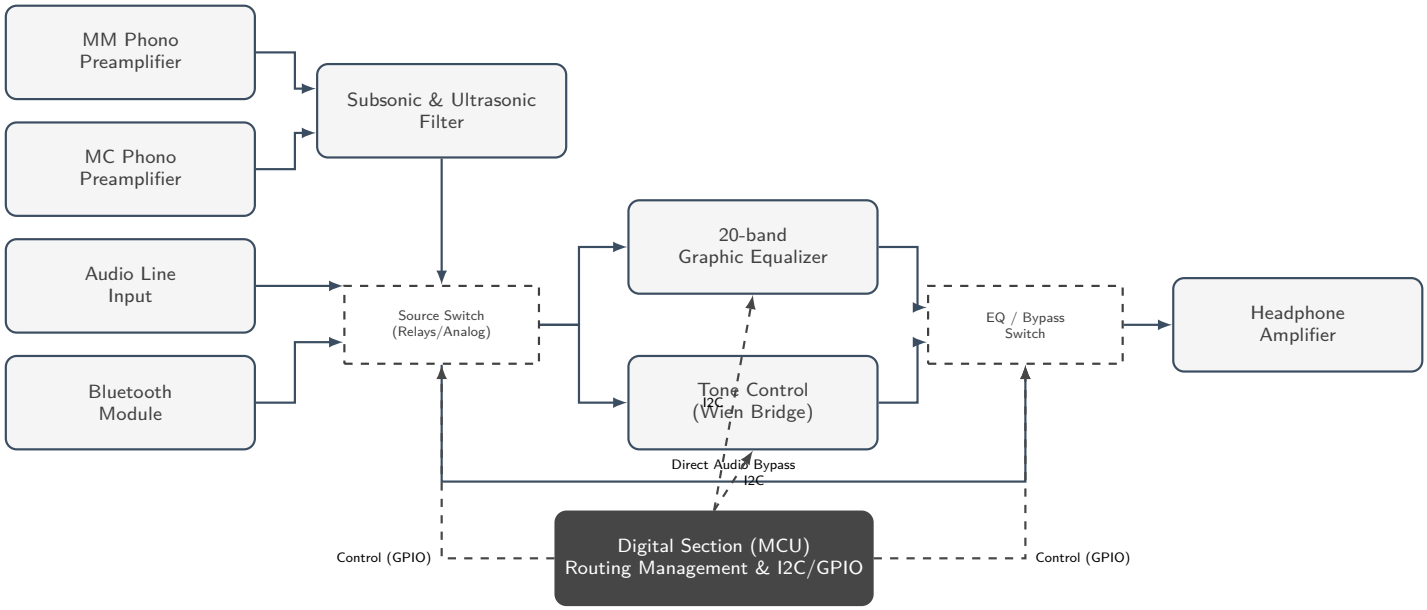
3D Model (Rear-1)



3D Model (Rear-2)

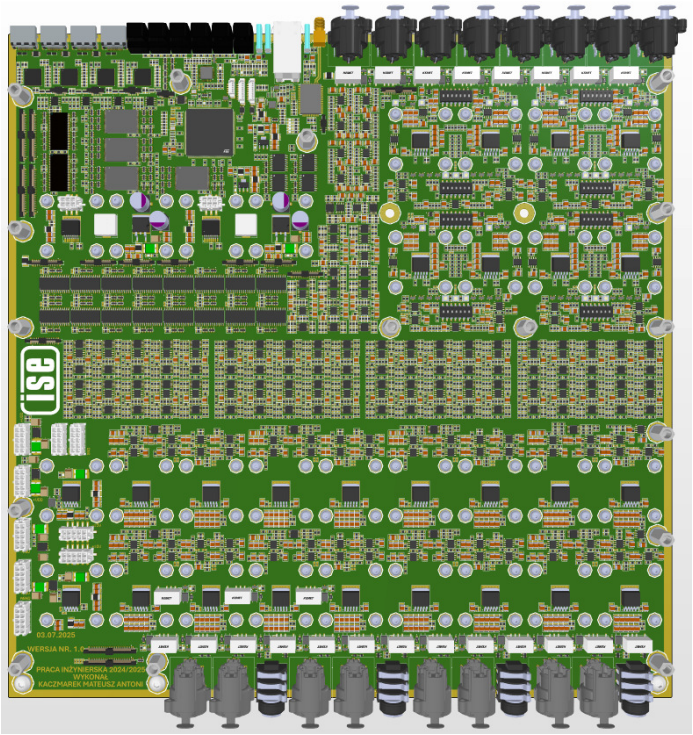
Subsystem: Main Audio Processing Board (Mixed-Signal PCB)

Analog-to-Digital Processing (Mixed-Signal Design): The designed mainboard is a massive printed circuit board constituting a hybrid of analog and digital technology. Due to the presence of sensitive acoustic signals, a key design challenge was the rigorous separation of power domains: the analog section is powered by a symmetrical ± 15V voltage, while digital blocks operate on independent 5V and 3.3V lines. The board integrates multiple advanced functional blocks, providing enormous audio path configuration capabilities.

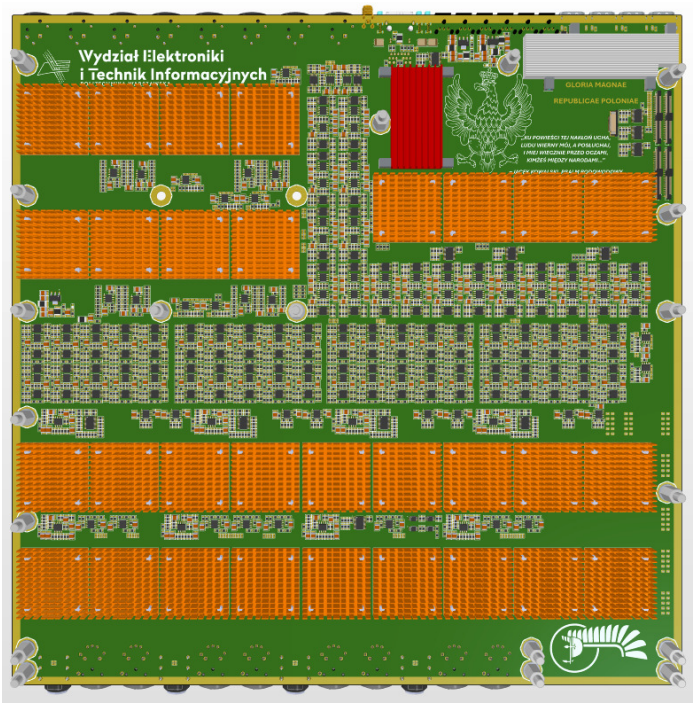


Topology and Functionality of the Audio Path: The board is equipped with dedicated, low-noise preamplifiers for MM (Moving Magnet) and MC (Moving Coil) phono cartridges. The signal then passes through a hardware subsonic and ultrasonic filter protecting the transducers from unwanted oscillations. Signal source switching (phono inputs, line inputs, wireless Bluetooth module) is implemented using relays and dedicated analog switches controlled by the microcontroller. Subsequently, the signal can be processed in a 20-band graphic equalizer and/or a precise Wien bridge tone control. The architecture allows bypassing the EQ section entirely (Bypass) to preserve absolute signal fidelity.

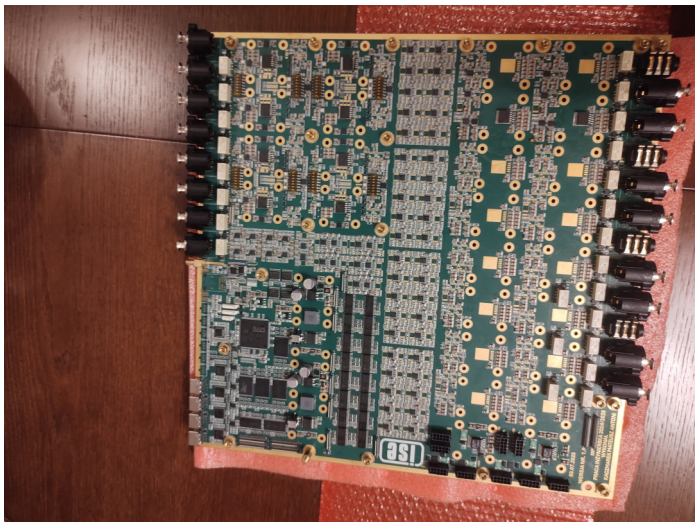
Hardware Implementation: The project was successfully transitioned from the simulation phase to physical implementation. Due to the packaging density of thousands of passive components (forming filters and EQs), the PCB has significant dimensions. The assembled and booted mainboard proves the effectiveness of the applied design rules regarding crosstalk reduction and maintaining a high Signal-to-Noise Ratio (SNR). The operation of power supply blocks, communication interfaces (Bluetooth), and key acoustic amplification paths was successfully verified.



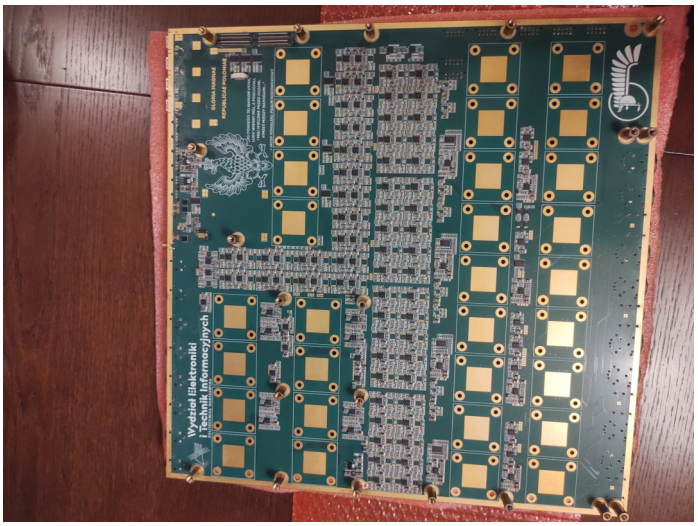
3D PCB Model (Top) - Dense layout of analog blocks



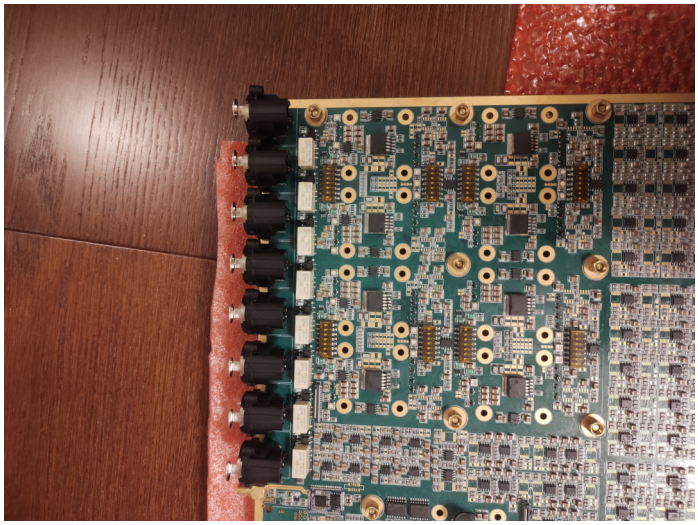
3D PCB Model (Bottom) - Graphic EQ section and heatsinks



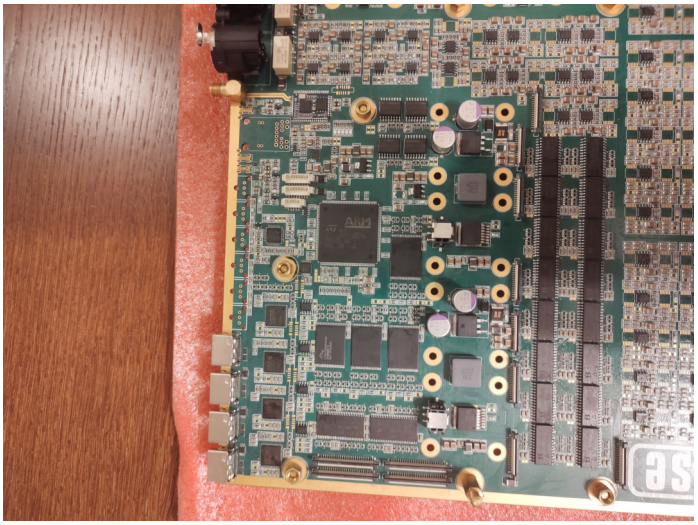
Manufactured main board (Top)



Assembled bottom section (Bottom)



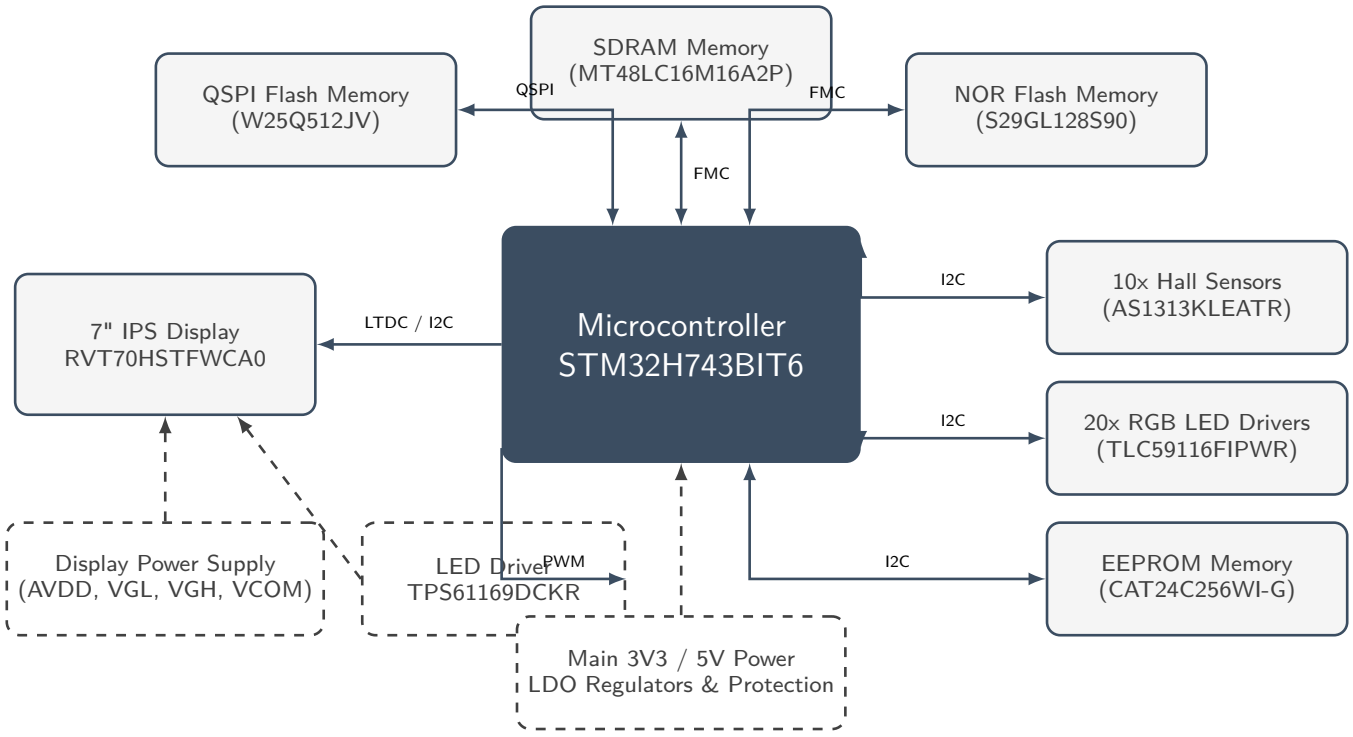
Close-up on low-noise MM and MC input paths



Digital section with master microcontroller and Bluetooth module

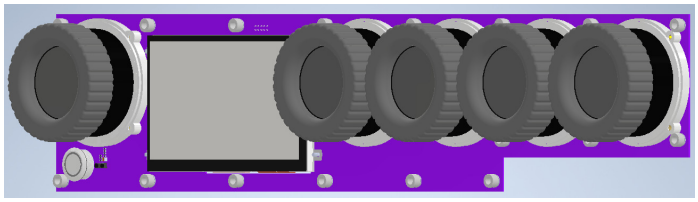
Subsystem: Front Panel and User Interface (HMI)

Control System Architecture (HMI): The front panel acts as an advanced Human-Machine Interface (HMI) based on the powerful STM32H743BIT6 microcontroller. This unit manages a complex peripheral system, including a 7-inch IPS touch screen, communicating via a 24-bit hardware LTDC interface. To ensure fluid operation of the Graphical User Interface (GUI) based on the TouchGFX framework, the system is augmented with external SDRAM (frame buffering) connected via a parallel FMC bus, and fast NOR Flash memories (QSPI and FMC interfaces) for storing graphical assets and code.



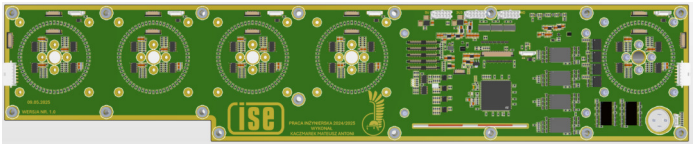
Contactless Magnetic Encoders and Power Supply: A key solution in this project is the complete elimination of classic mechanical potentiometers. The panel is equipped with five rotary knobs featuring their own displays. Their rotation is read contactlessly by a matrix of Hall effect sensors (AS1313KLEATR-2000). The sensors react to magnetic field changes from neodymium magnets placed with alternating poles, allowing the microcontroller to precisely determine the rotation vector and speed. Each knob is surrounded by an RGB LED ring controlled by dedicated TLC59116FIPWR chips. Powering such a complex system required designing a multi-voltage DC/DC section on the board, generating stable voltages for the IPS matrix (AVDD 9.8V, VCOM 3.8V) and biasing voltages VGH (18V) and VGL (-6V) using dedicated charge pumps and regulators.

Mechanical Design and GUI (TouchGFX): The designed 6-layer PCB posed a significant technological challenge due to its dimensions (656×130 mm). Because of the large span of the laminate and the mechanical pressure exerted during knob operation, a dedicated stiffening plate was applied to the bottom side during assembly. It was laser-cut from stainless steel, ensuring structural rigidity. The device's graphical interface was written using the TouchGFX framework, which enables smooth animation of the hardware-calculated audio spectrum. Fast Fourier Transform (FFT) is performed using the optimized CMSIS DSP mathematical library, allowing real-time analysis.

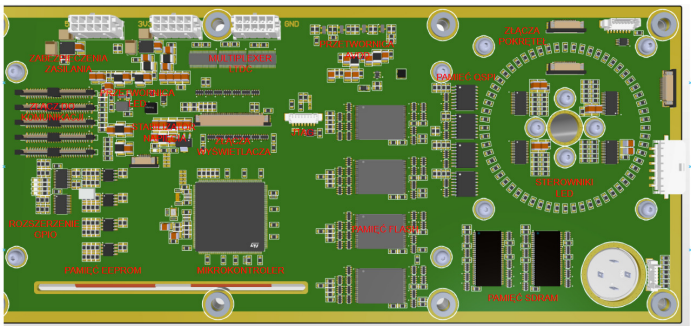


3D Model - Placement of the 7" display and 5 knobs

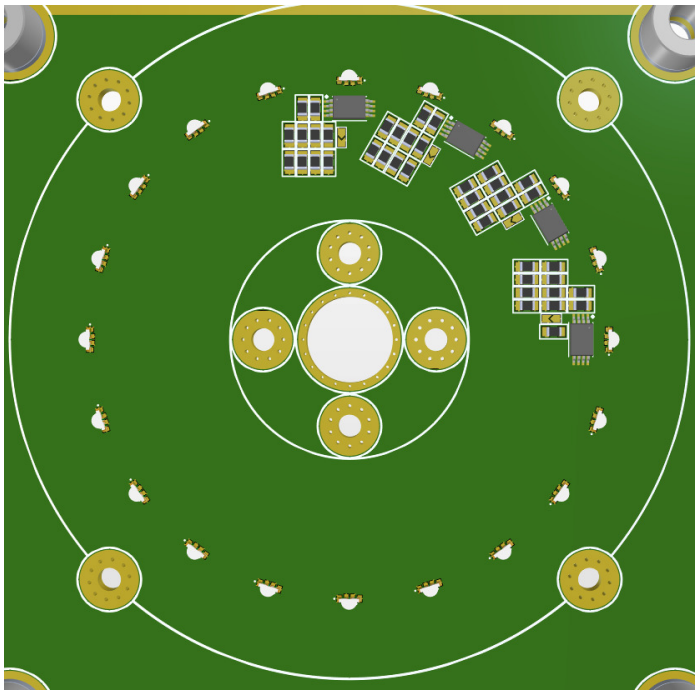
PCB View (Top)



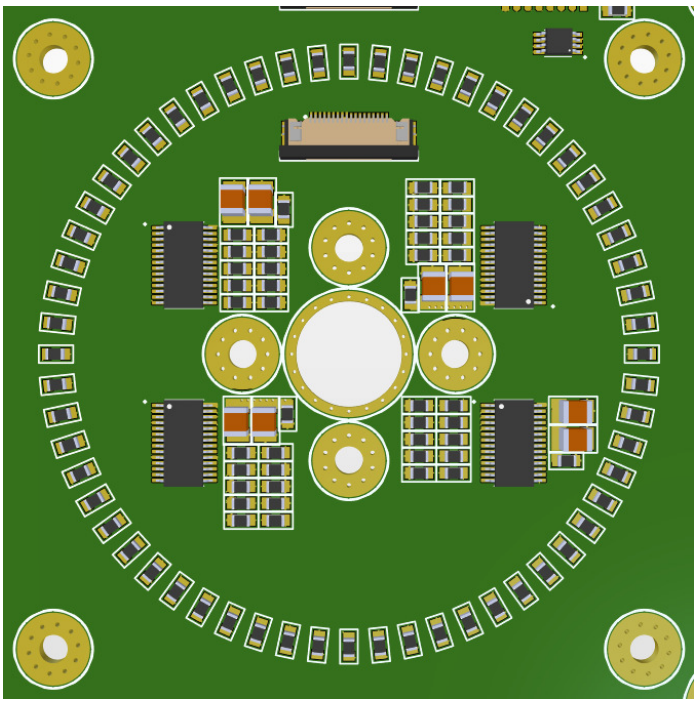
PCB View (Bottom)



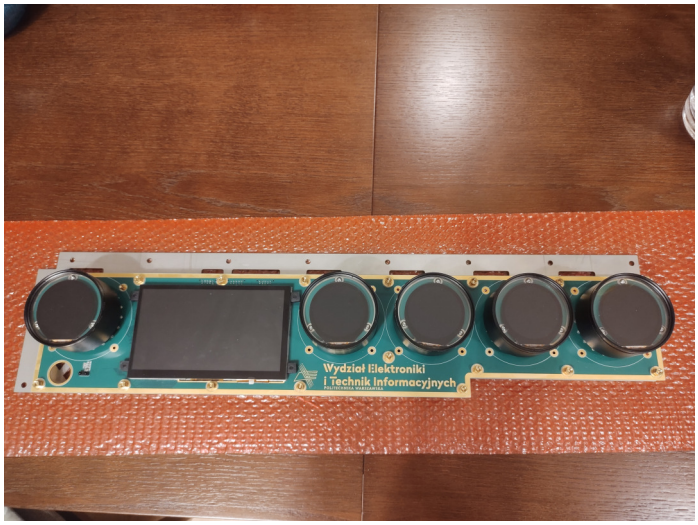
Routing details of the central section (MCU and SDRAM)



Layout detail - Matrix of magnetic Hall sensors

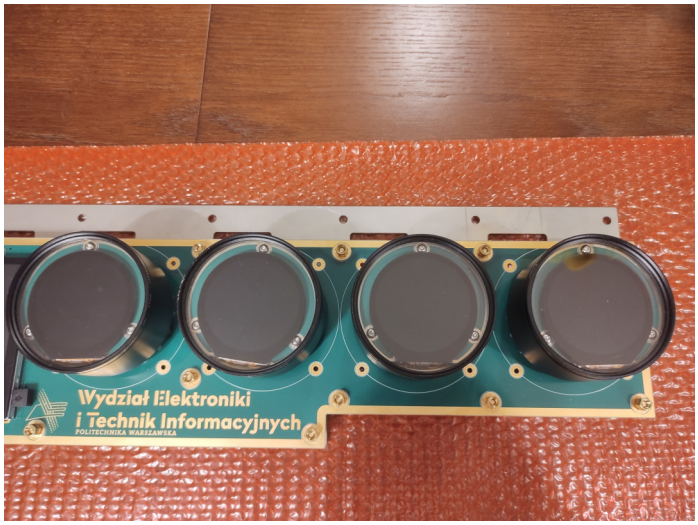


Layout detail - LED drivers (I2C)

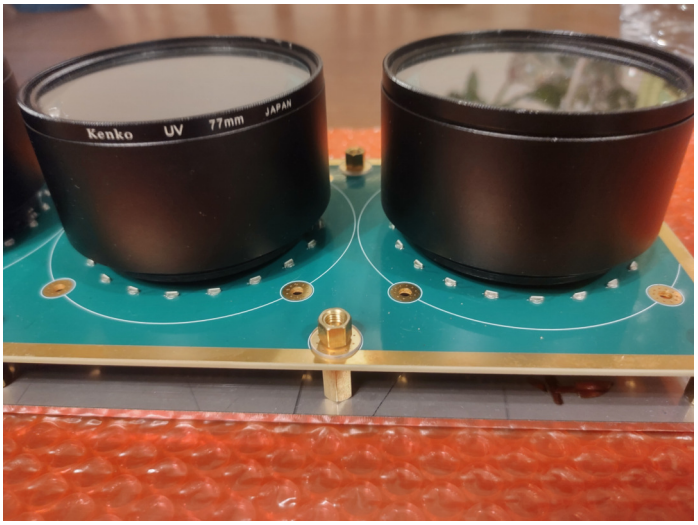


Assembled panel (Front) - wide shot with the display matrix

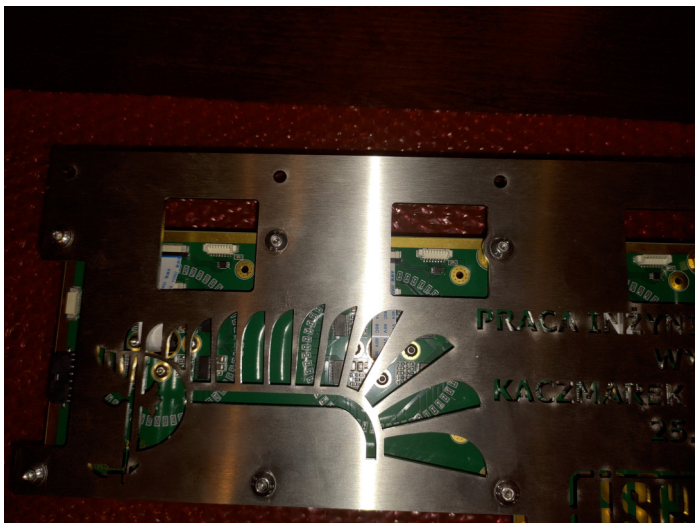
Assembled panel (Front) - close-up on the 7" display



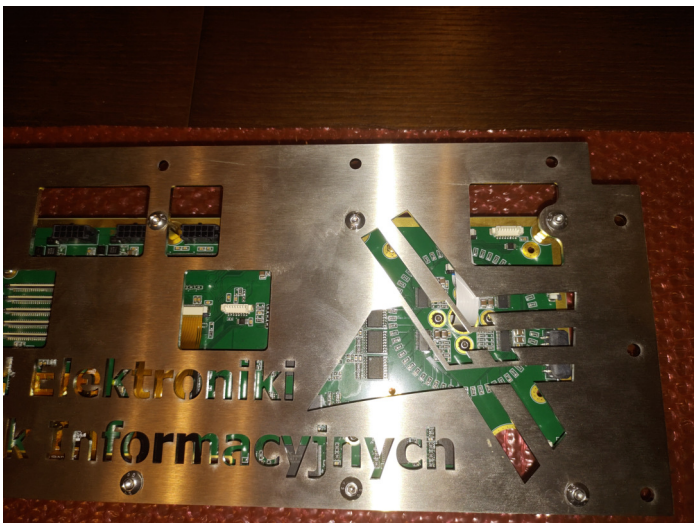
Right section of magnetic knobs (rotary encoders)



Close-up on encoders equipped with bottom LEDs



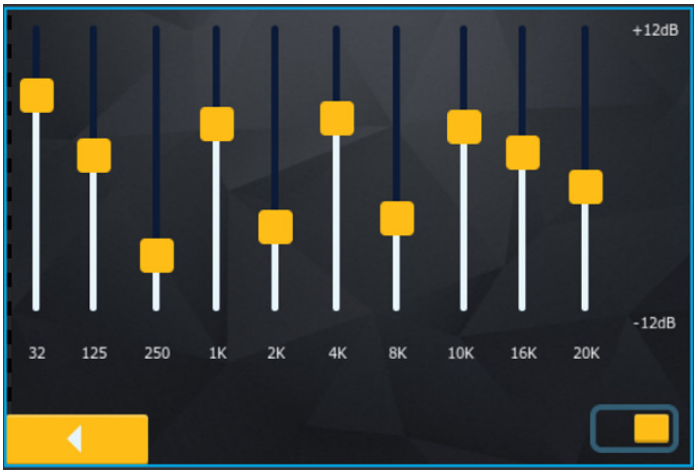
Stainless steel stiffening plate



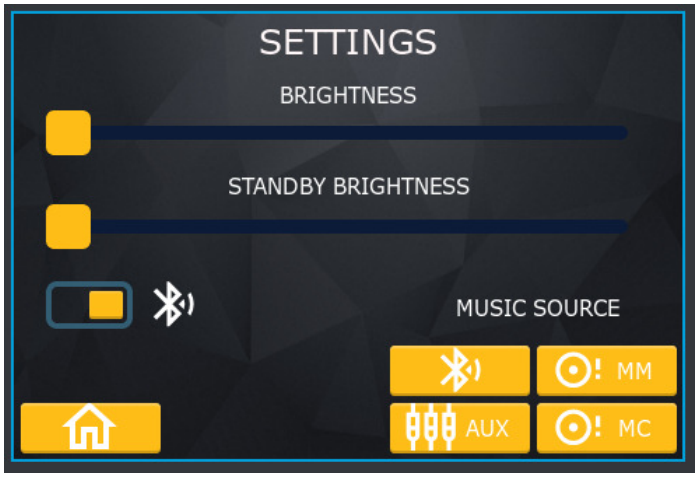
Laser cutting details and access to communication connectors

Full view of the steel mechanics on the rear of the panel (Bottom)

GUI - Main screen with audio spectrum visualization (FFT from CMSIS)

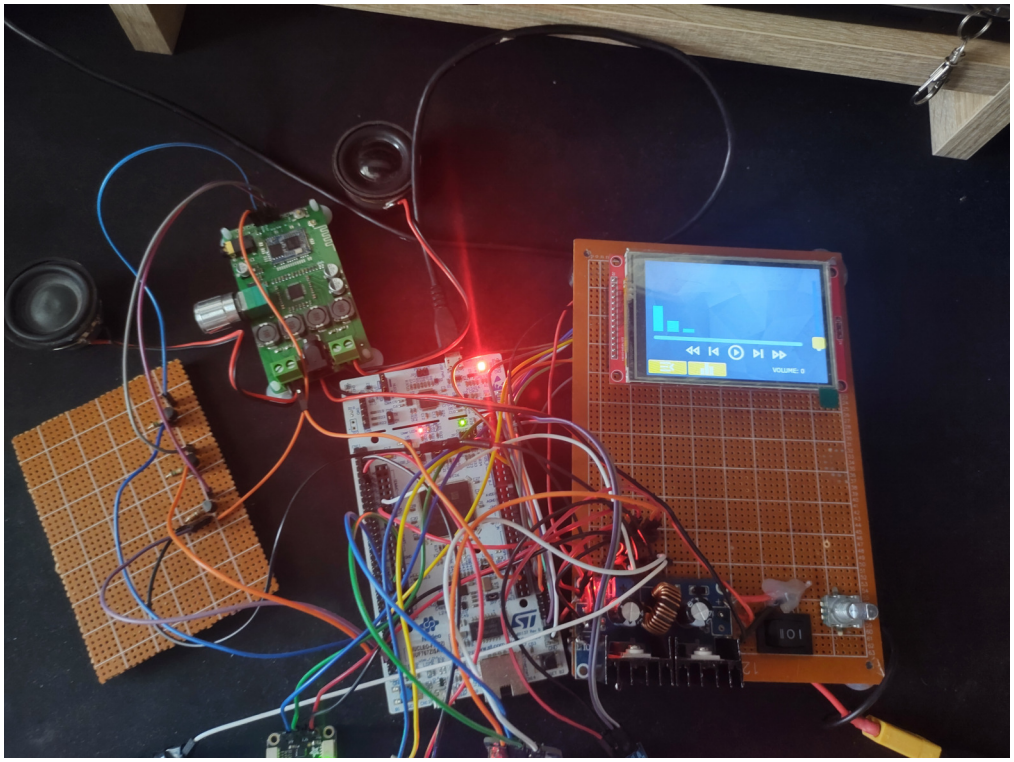


GUI - Graphic band equalizer control



GUI - Settings panel (Display brightness, Bluetooth module)

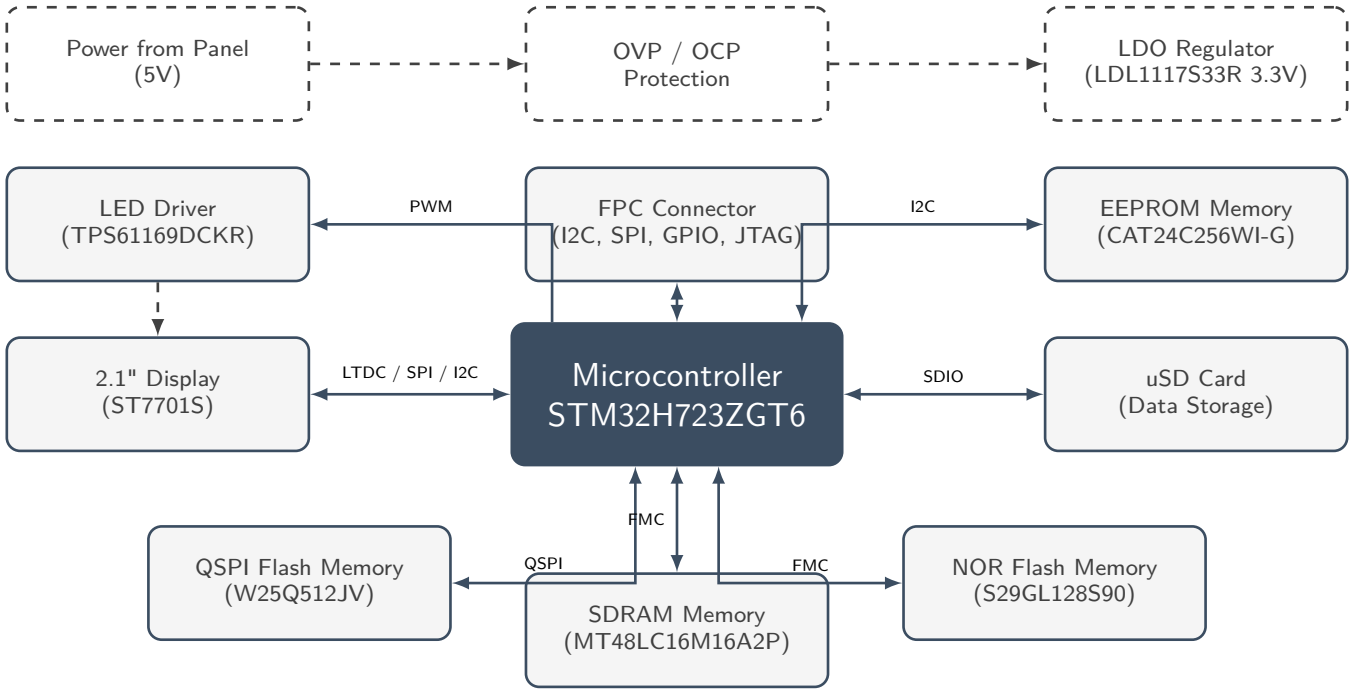
GUI - Minimalist start screen (Standby Mode)



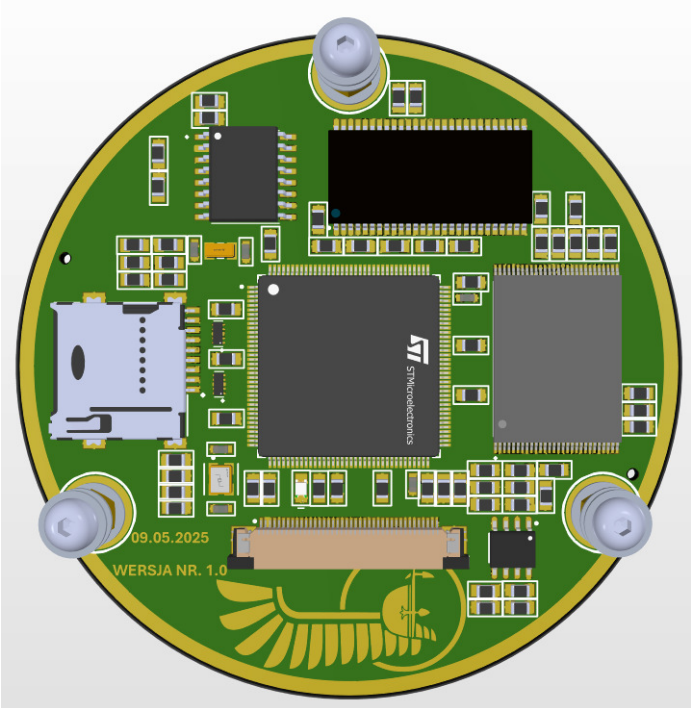
Prototype

Subsystem: Smart Knob with Display

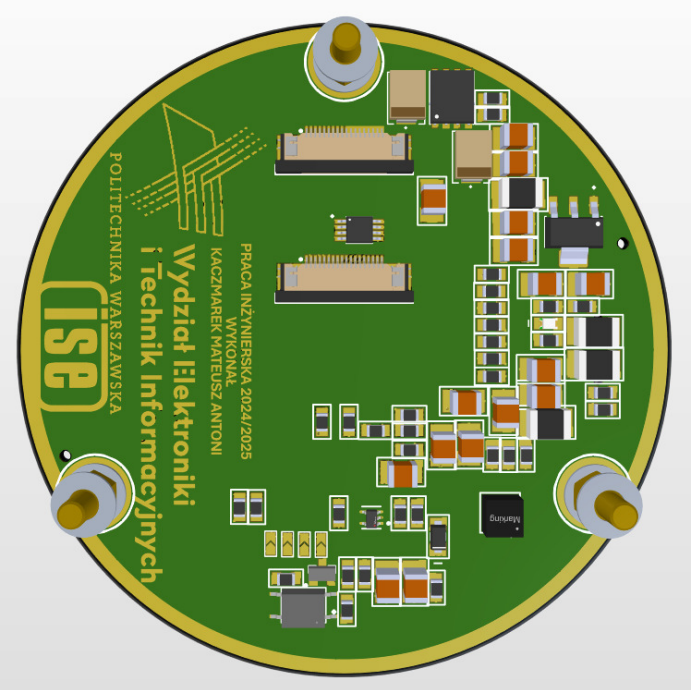
System Architecture and Display Management: The smart knob is a highly integrated, autonomous module acting as an advanced user interface. The central computing unit is the STM32H723ZGT6 microcontroller, responsible for rendering graphics and communicating with the main panel. Parameter visualization is performed on a circular 2.1-inch display (ST7701S driver), handled via an 18-bit parallel video bus (LTDC). To meet the demands of a fluid interface (TouchGFX), the system is expanded with external SDRAM memory (FMC bus), QSPI Flash memory for graphical assets, a NOR Flash chip, and a microSD card slot. The entire unit communicates with the front panel motherboard via a flexible FPC ribbon cable (SPI, I2C, JTAG, GPIO interfaces).



High-Speed PCB Design: The electronic circuit was laid out on a 6-layer printed circuit board in the shape of a circle with a diameter of 76 mm (FR-4 laminate, $T_g = 155°C$, ENIG finish). Special care was taken regarding impedance control and precise length tuning (meandering) for address and data lines of high-speed SDRAM and Flash memories, as well as the parallel LTDC video bus. The screen backlight is controlled by a PWM signal via the integrated TPS61169DCKR converter, powering a cascade of white LEDs (12.8V, 20mA).



3D PCB Model (Top) - MCU, FPC connector, memories, and uSD reader

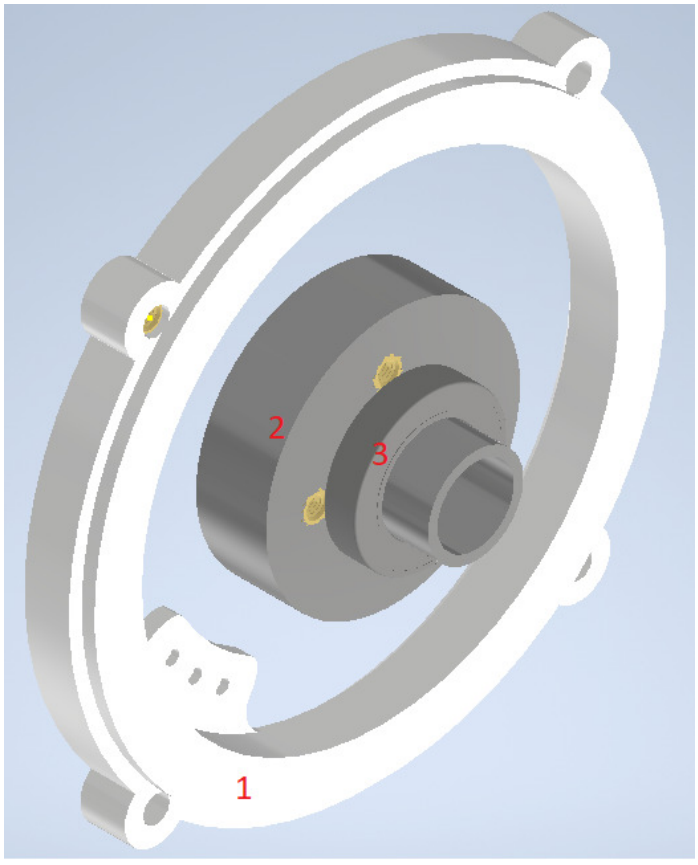


3D PCB Model (Bottom) - Power supply section

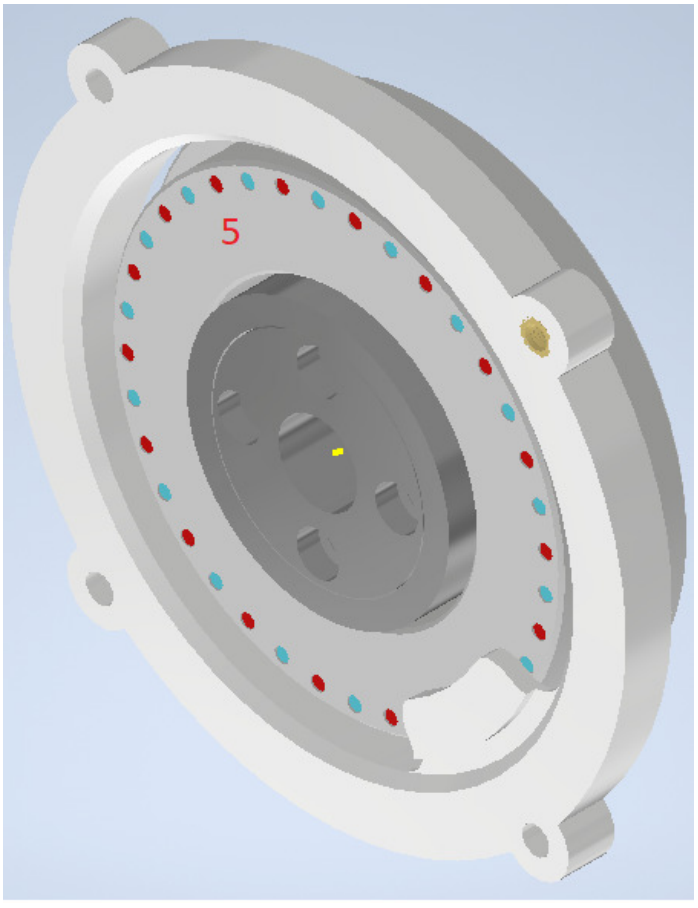
Manufactured board (Top) - Front with the 2.1" matrix applied

Assembled board (Bottom) with a flexible FPC signal ribbon

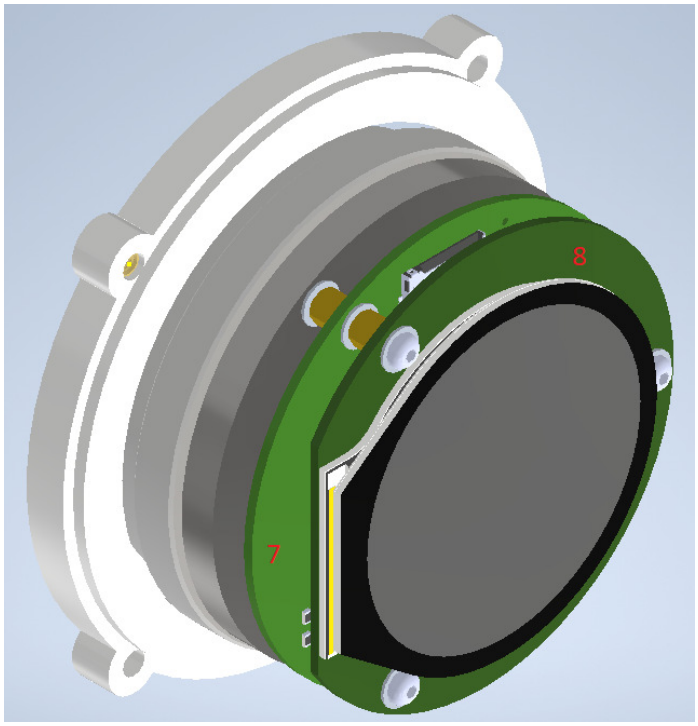
Integrated Mechanical and Optical Design: The knob is an electromechanical system consisting of 11 precision parts, combining various material processing technologies. The base of the module is a transparent cover printed from photopolymer resin (SLA), allowing light from RGB LEDs located on the main panel to pass through. The axis of rotation is a CNC-machined steel sleeve integrated with a ball bearing (6802). The rotor, made of PLA plastic (FDM printing), has neodymium magnets glued in alternately. They rotate just above the Hall sensors on the mainboard, enabling contactless position and rotation speed readings. The top layer is an anodized aluminum housing finished with protective glass and a flexible TPU elastomer grip.



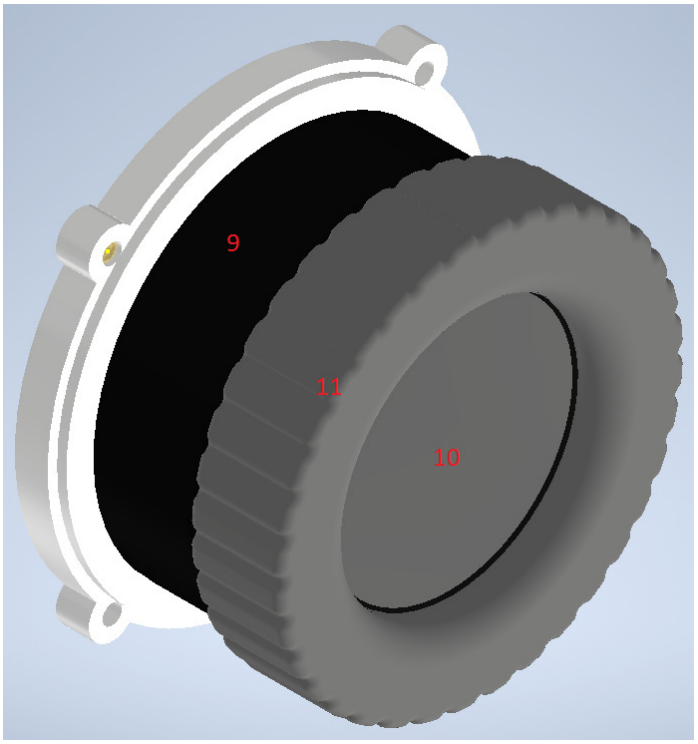
Base - SLA Cover, CNC sleeve, and bearing



Alternating neodymium magnets arrangement in the rotor



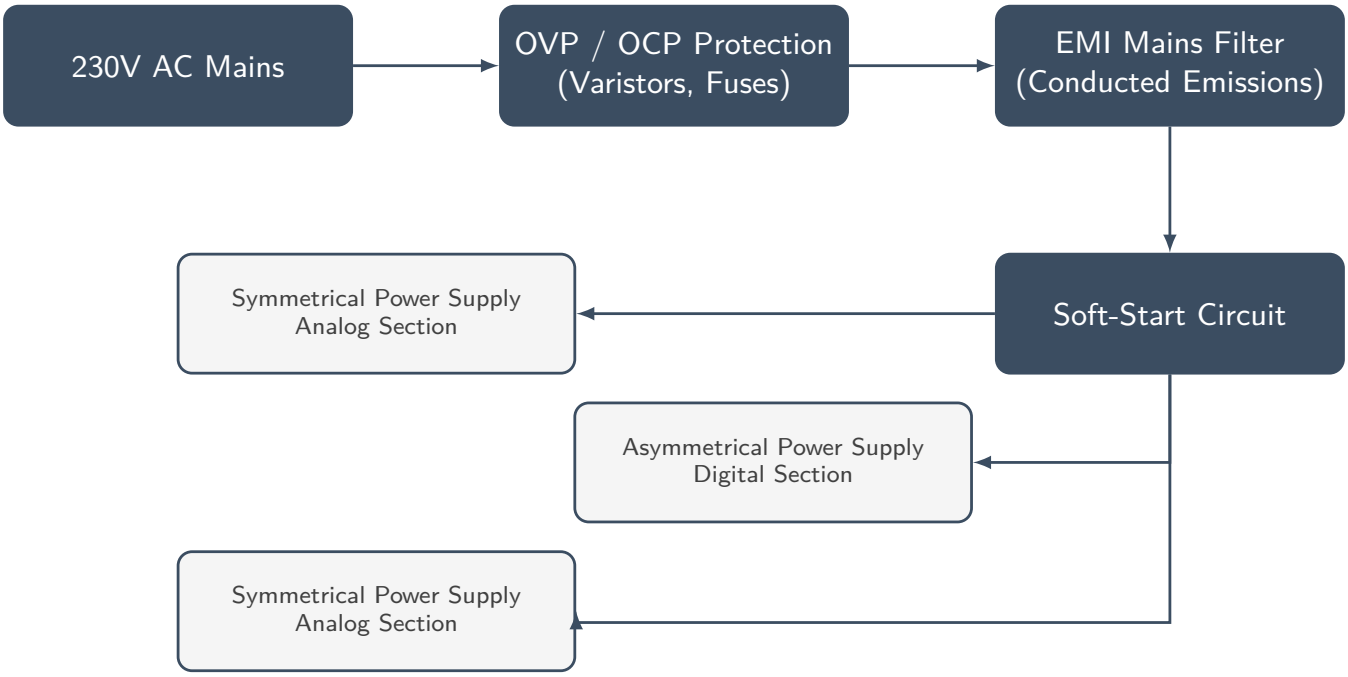
Assembly of the integrated PCB and the round 2.1" screen



Complete module - Aluminum sleeve, glass, and TPU cap

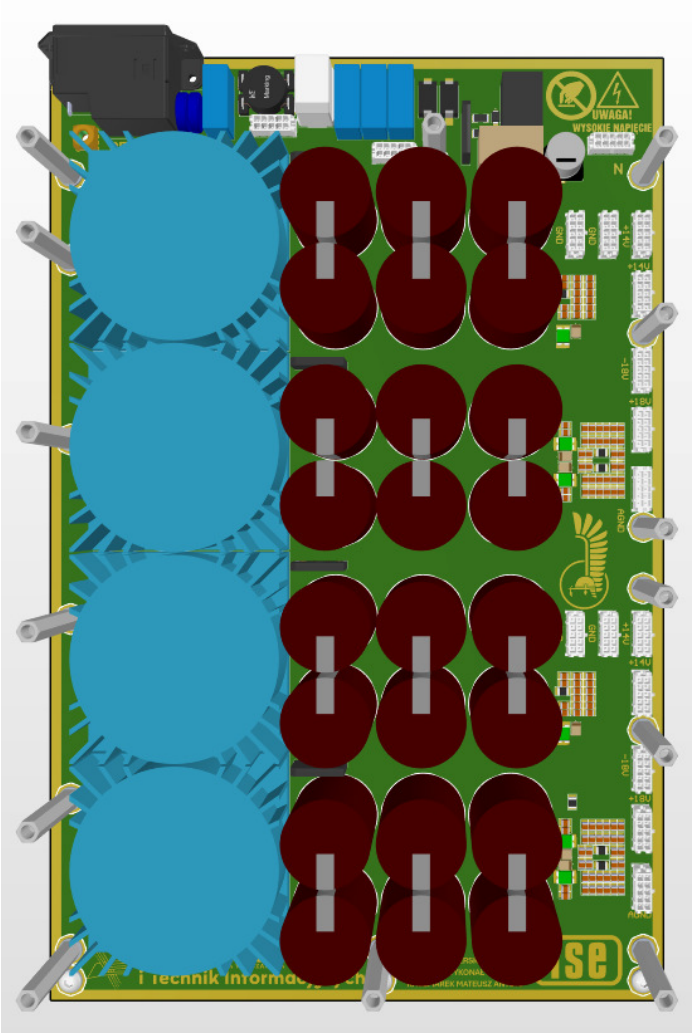
Subsystem: Multi-transformer Linear Power Supply

Power Architecture and Energy Distribution: The audio system's power supply was designed as a multi-section linear circuit. The main objective was to provide absolutely stable voltage and full galvanic isolation between sensitive analog paths and the digital control section. The system utilizes up to four dedicated toroidal transformers. Their selection was dictated by very low magnetic stray fields and high efficiency, a critical parameter near highly sensitive phono preamplifiers (RIAA). The analog path is powered by symmetrical sections (50VA transformers, bridge rectifiers, massive filter capacitor banks), while the digital part is powered by an independent transformer with series-connected windings.

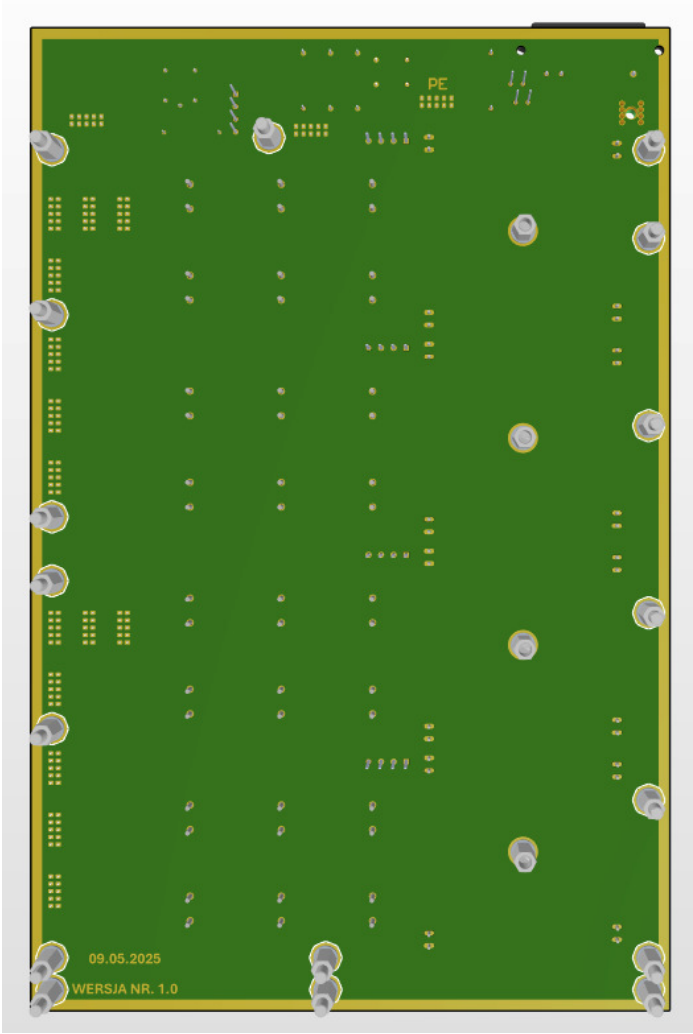


EMI Filtering and Safety (OVP/OCP): To comply with stringent Electromagnetic Compatibility (EMC) standards, filters limiting the emission and reception of conducted interference were implemented at the 230V mains input. Filtration is based on dedicated Class X safety capacitors (multiple suppression of differential-mode noise) and Class Y capacitors, combined with common-mode and differential-mode chokes. Y capacitors drain asymmetrical interference directly to the protective earth (PE) conductor, which is galvanically bonded to the device's metal enclosure. Overvoltage protection is provided by varistors, and overcurrent protection by fuses near the IEC connector.

Soft-Start Circuit and PCB Design: The use of four toroidal transformers and the massive capacitance of the filtering banks generates an extremely high inrush current upon powering on the device, which could trip electrical installation breakers. To prevent this, a hardware "Soft-Start" circuit was implemented. It utilizes ceramic power resistors (100Ω, 5W) that limit the inrush current for approximately the first second. Subsequently, an analog timer circuit (based on a bridge rectifier and an RC time constant) shorts the resistors using a relay. The entire power supply was built on a massive, 4-layer printed circuit board measuring 400×260 mm (FR-4 laminate, $T_g = 155°$C, ENIG), which allowed for the safe routing of high-current traces while maintaining proper insulation clearances.



3D PCB Model (Top) - Massive capacitor banks and filters



3D PCB Model (Bottom)

Academic and Student Projects

VGA SoC Video Controller (Zynq-7000 FPGA)

10.2025 - 02.2026 (Completed)

Digital IC Designer | Academic Project

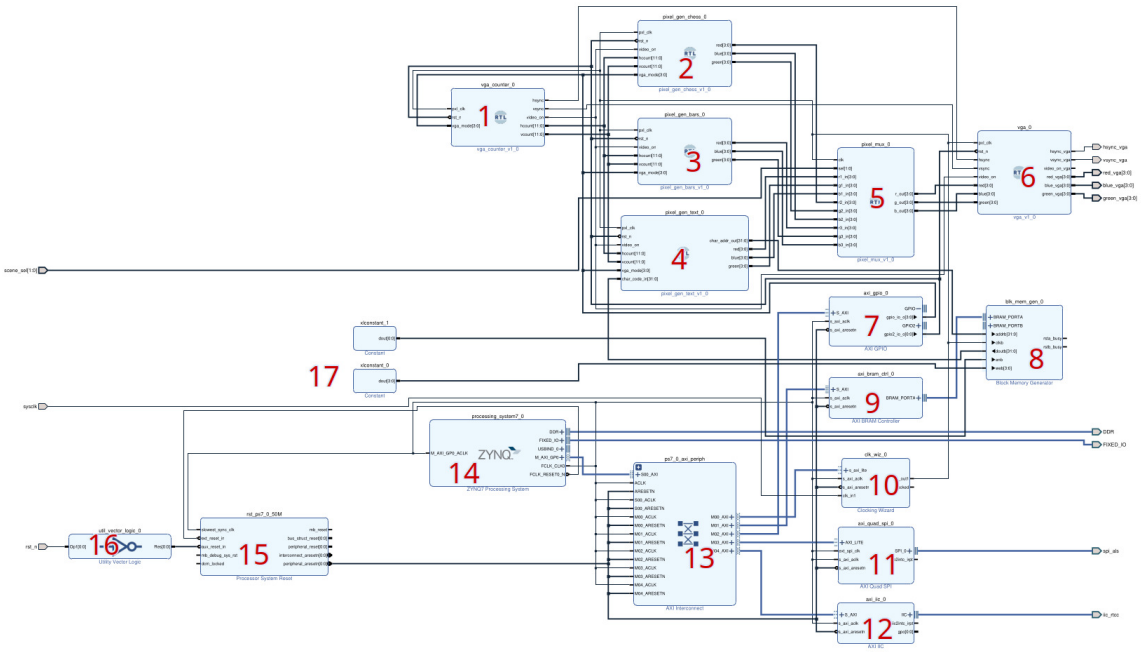
Design and physical implementation of a heterogeneous embedded System on a Chip (SoC) on the Digilent Zybo Z7-10 platform. Integration of a custom VGA video controller written in VHDL (Programmable Logic - PL) with an ARM Cortex-A9 processor (Processing System - PS) via the AXI4 bus was achieved. In a bare-metal model, the system acquires data from peripherals (I2C, SPI) and renders a dynamic graphical interface in real-time.

Key Hardware and Software Solutions:

- Dynamic Reconfiguration Port (DRP): A tunable Phase-Locked Loop (MMCM/PLL) enables smooth, real-time switching of VESA standards (from 640x480 to 1024x768 @ 85Hz) from the processor level. Direct Register Access to AXI4-Lite was utilized, bypassing HAL libraries for complete time determinism.
- Pipeline Architecture: A dynamic text generator acting as a DMA controller. A 4-stage processing architecture with a 3-stage delay line was applied, effectively compensating for cascading read delays from Video RAM (BRAM) and character memory (Font ROM).
- Clock Domain and Reset Management: Separation of asynchronous clock domains (System Clock vs. Pixel Clock up to 100 MHz) using hardware True Dual Port RAM. An independent, asynchronous reset system (Active-Low) for the video domain was implemented, guaranteeing immunity to PLL instabilities.
- Bare-metal Software (C): Custom interface drivers (I2C, SPI) operating directly on registers and a hardware Watchdog Timer protecting the AXI bus from stalling. The complete application is launched from QSPI Flash memory using the FSBL bootloader.

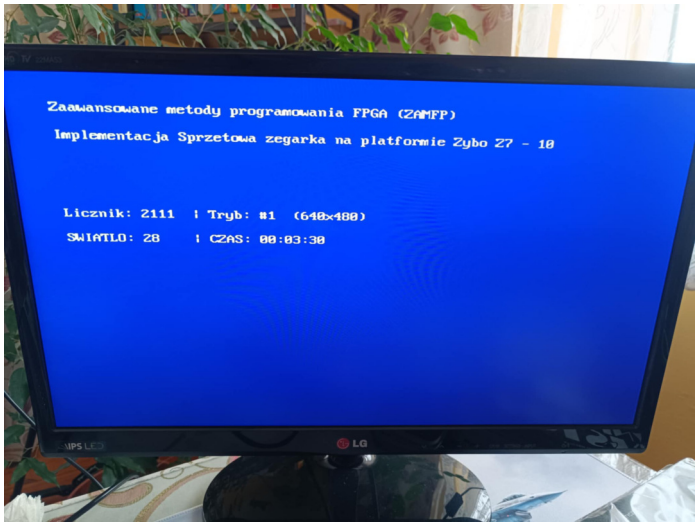
System Architecture (Hardware Block Design): The SoC structure relies on the tight integration of custom RTL modules with Xilinx IPs. Main functional blocks are numbered on the diagram below:

- [14, 13] Processing System: A dual-core ARM Cortex-A9 processor (Zynq PS) communicating with the PL logic through a central AXI Interconnect arbitration block.
- [8, 9] Frame Buffer (VRAM): Block Memory Generator (True Dual Port) acting as a clock domain isolator, controlled from the processor by an AXI BRAM controller.
- [10, 15, 16] Clock and Reset Management: A Clocking Wizard with a DRP port generating the pixel clock signal, and a Processor System Reset block distributing asynchronous clear signals (along with a polarity inversion gate).
- [1-6] Video Pipeline (VHDL RTL): Custom video logic: Vga Counter (timebase), content generators (Checkerboard, Bars, Text), a Pixel Mux signal multiplexer, and a VGA Output module.
- [7, 11, 12] AXI Peripherals: Hardware controllers: GPIO block (triggering modes and soft-reset), AXI Quad SPI (handling the ALS light sensor), and AXI IIC (communication with the RTCC clock).

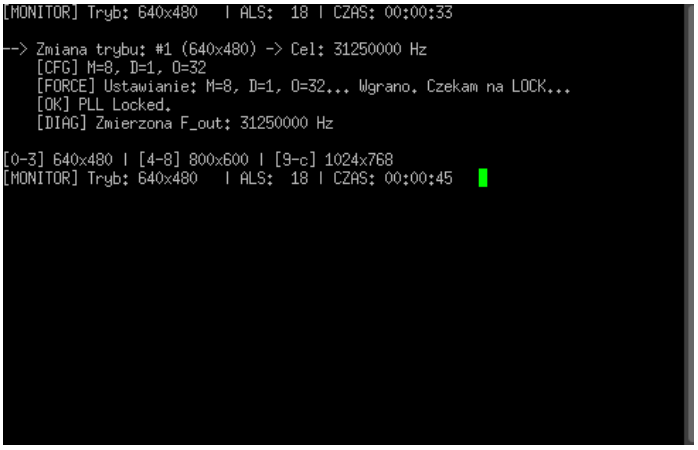


SoC block architecture with numbered functional modules (Vivado 2023.2)

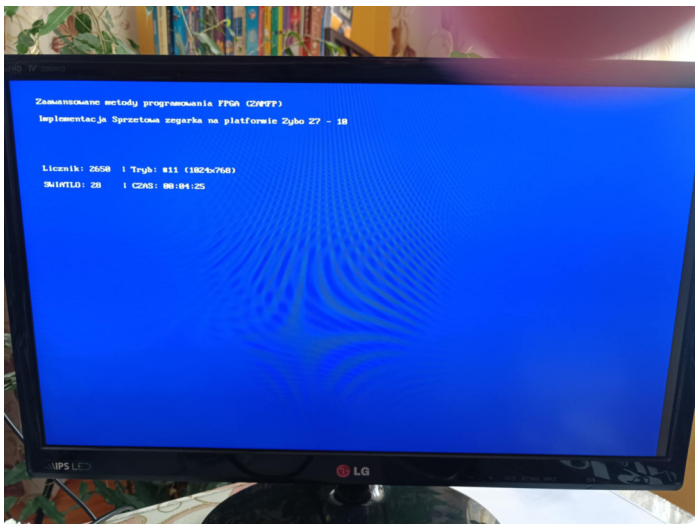
Physical Verification and Resolution Tests:



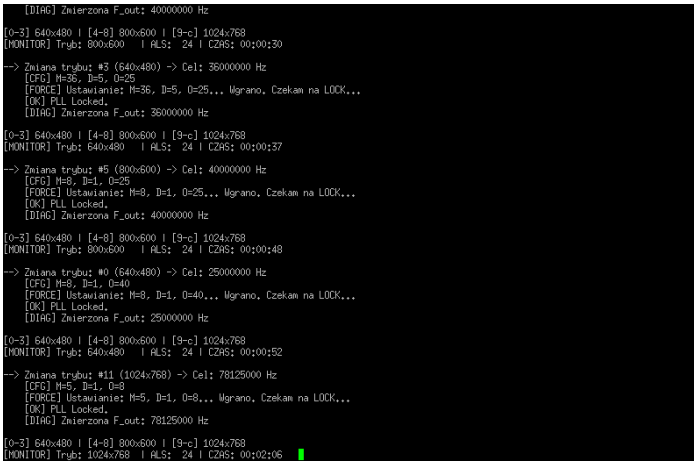
Hardware-generated image with peripheral readings (640x480 @ 72Hz Mode)



UART logs confirming hardware PLL lock (LOCK) without FPGA reloading



System response to forcing 1024x768 @ 75Hz resolution (highest supported by monitor)



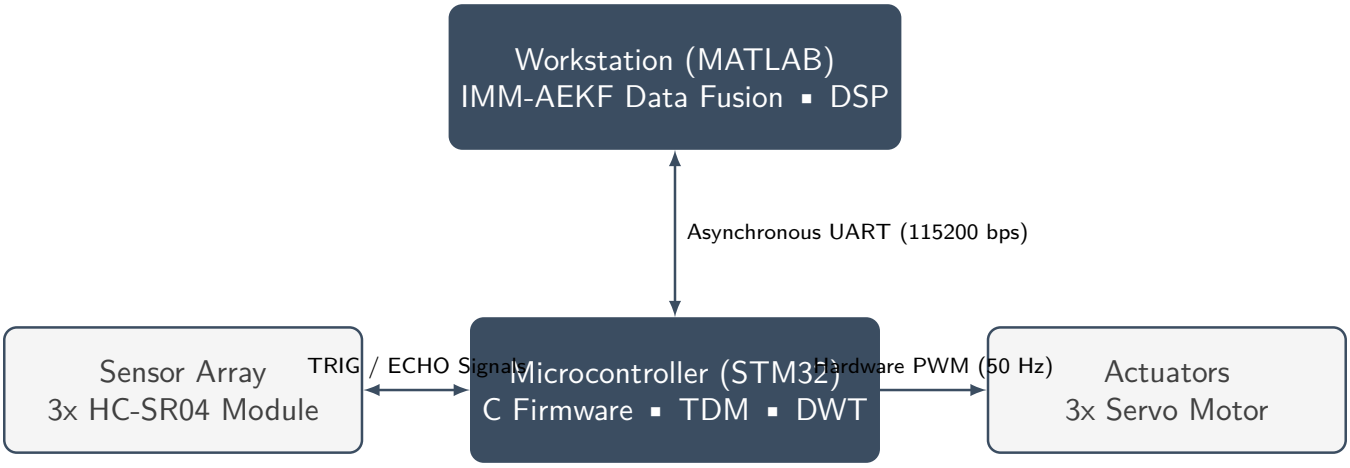
Diagnostics confirming on-the-fly M, D, O register retuning for ∼78 MHz clock

Adaptive Sonar System (SRMP)

02.2026 - 06.2026 (Completed)

Systems Engineer | Independent Academic Project

Design, hardware implementation, and analysis of an advanced sonar system capable of precise real-time object tracking. The device integrates a custom ultrasonic sensor array, ARM Cortex-M microcontroller software, and stochastic data fusion (EKF, IMM) in the MATLAB environment.



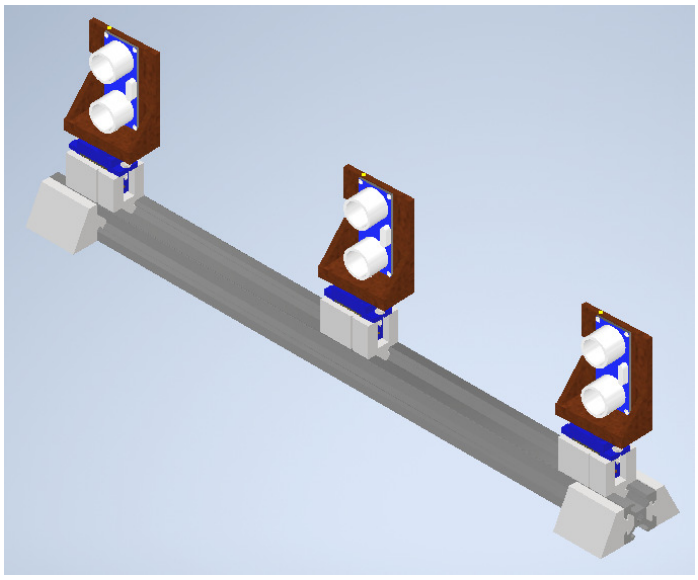
Key Technological Solutions and Architecture:

- IMM-AEKF Estimator: An Asynchronous Extended Kalman Filter with Interacting Multiple Model architecture (Markov chains) was implemented, smoothly switching between a dormant (CV) and maneuvering (CA) model.
- Clutter Mapping: A spatial echo gating algorithm was created during the calibration phase, effectively rejecting reflections from fixed environmental obstacles.
- Feedback Loop and Defocusing: A feedback loop was programmed based on the trace of the covariance matrix. An increase in system uncertainty forces an expansion of the sensors' viewing angle.
- Embedded C Firmware (STM32): The hardware DWT register was utilized for microsecond-level time measurement. Measurements were time-division multiplexed (TDM) with an adaptive PRI interval.

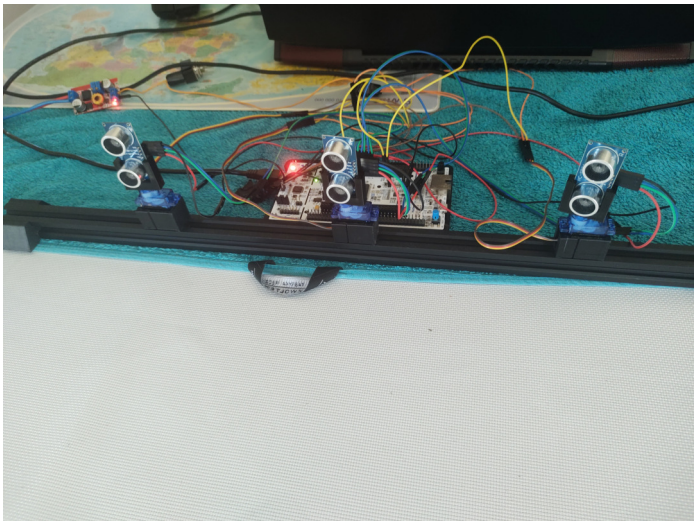
Results and Metrological Verification:

- Static Measurements: The precision of the estimator was confirmed with a localization error of only a few millimeters relative to physical measurements.
- Stochastic Analysis: The Normalized Innovation Squared (NIS) indicator confirmed the correct mathematical tuning of the Q and R noise matrices.
- Dynamic Tracking: Maneuvering kinetics were successfully captured within calculated, extremely narrow $\pm 3\sigma$ confidence bands.

Mechanics and Electronic Integration: The hardware layer was realized based on a rigid frame made of V-Slot 2020 aluminum profiles. Connecting elements and mounts for ultrasonic transducers and servos were designed in a CAD environment and manufactured using 3D printing technology.

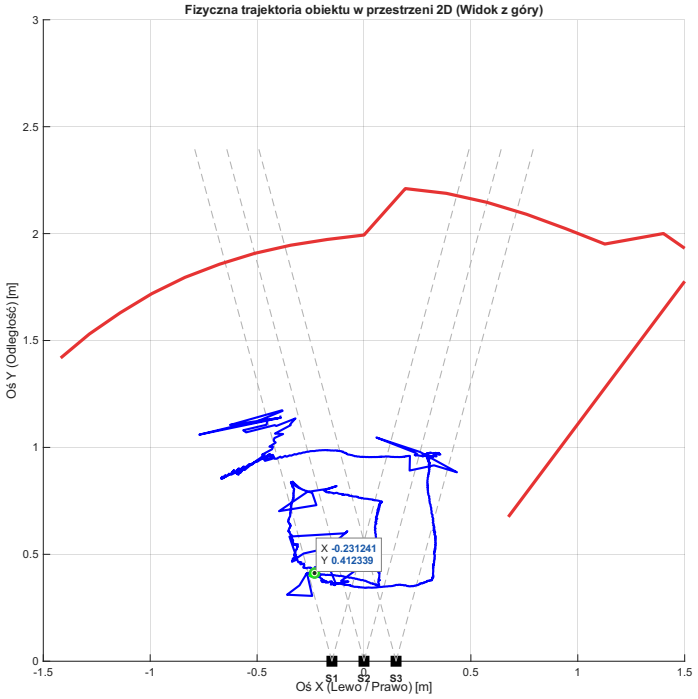


CAD model of the linear sensor array

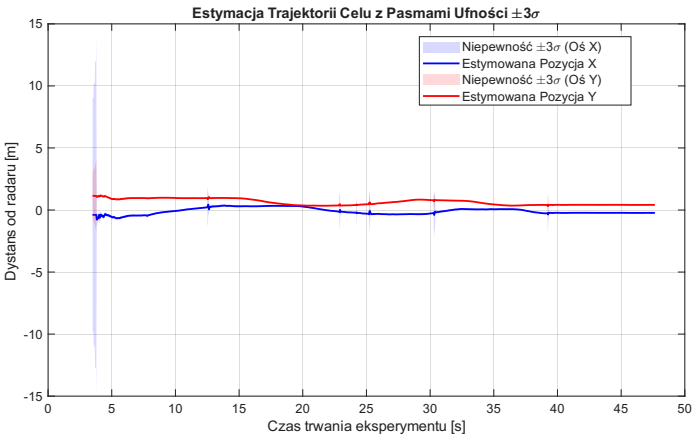


Physical realization of the STM32 testbed

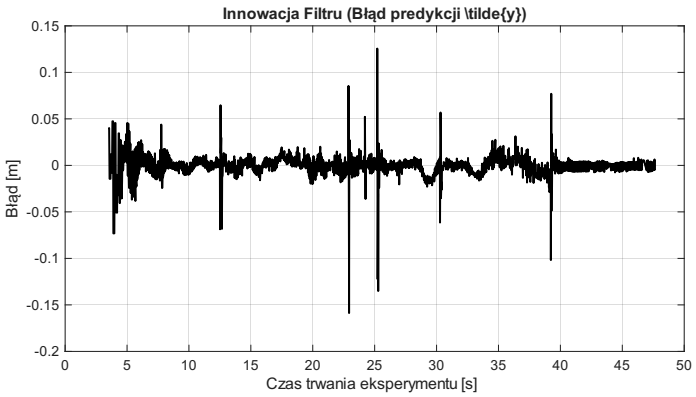
Estimates Visualization and Data Fusion: The analytical environment processes an asynchronous data stream from sensors in real-time. The estimator autonomously approximates radial velocity vectors and draws an environment map in 2D space.



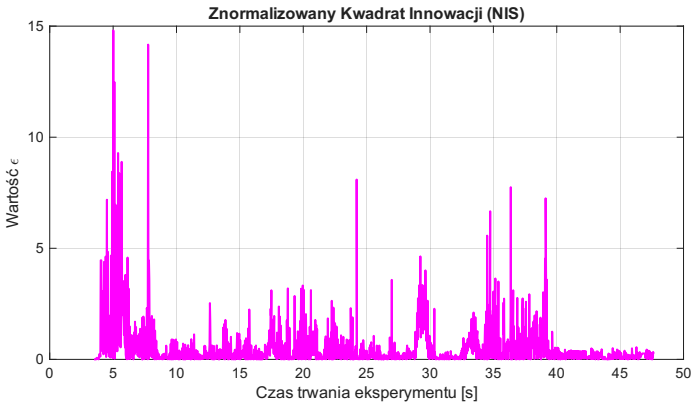
2D View: Estimation against the Clutter map



Trajectory estimation within $\pm 3\sigma$ bands



Filter Innovation (Prediction Error)



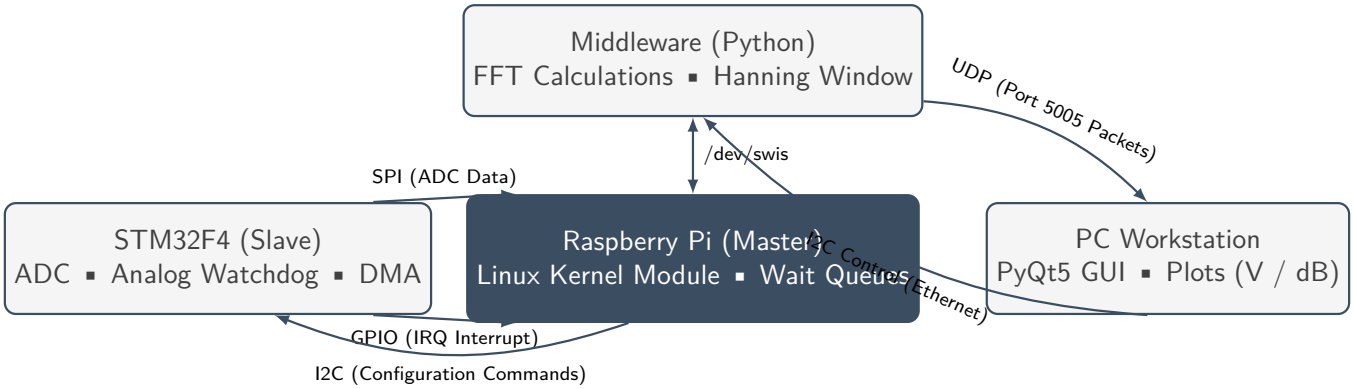
NIS Statistic (confirmation of Q and R tuning)

Data Acquisition System (STM32 + Raspberry Pi)

02.2026 - 06.2026 (Completed)

Co-developer (STM32 Firmware, DSP, GUI) | Academic Project

A project executed in a two-person team aimed at creating a system for high-speed data acquisition and analysis (a mini-oscilloscope and spectrum analyzer). The architecture relies on a Master-Slave setup. My scope of responsibilities included developing the microcontroller embedded software (STM32), writing signal analysis algorithms (FFT), and building a graphical PC application. My teammate was responsible for the Linux kernel layer on the Raspberry Pi (character device driver module).

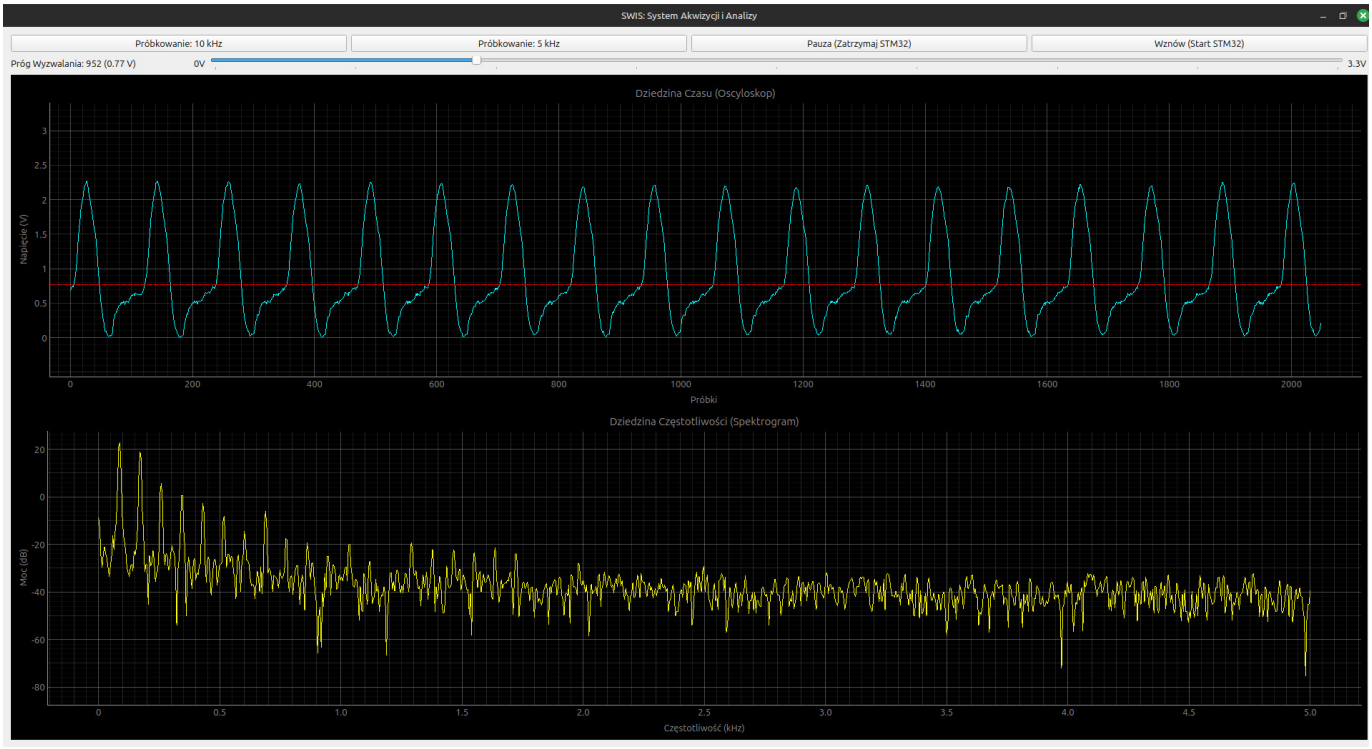


My Key Tasks and Implemented Solutions (Firmware & DSP):

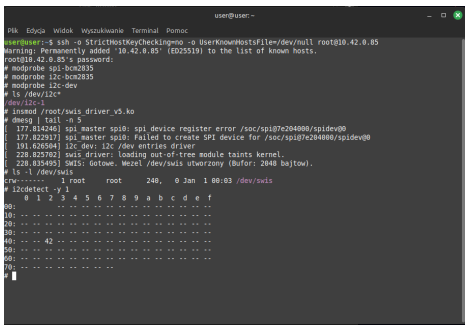
- Hardware Trigger (STM32): I implemented a hardware trigger function for the oscilloscope using the Analog Watchdog module. The IC monitors voltage and automatically triggers data collection upon exceeding a threshold, which is visible on the GUI as a stable signal start from the 0.77 V level.
- DMA Transfer and Optimization: I configured the TIM3 timer for rigid ADC clocking. Acquired 12-bit samples are compressed to 8 bits via bit-shifting and then sent in the background by the DMA controller over the SPI bus to the Raspberry Pi.
- Digital Signal Processing (DSP): I wrote a Python script running in the RPi user space that receives data from the character device file, removes the DC component, applies a Hanning window, and calculates the Fast Fourier Transform (FFT).
- User Interface (PyQt5): I developed a utility application with a GUI that converts FFT results logarithmically to decibels (dB) and scales ADC values to physical signal voltage in Volts.

System Integration and Test Results:

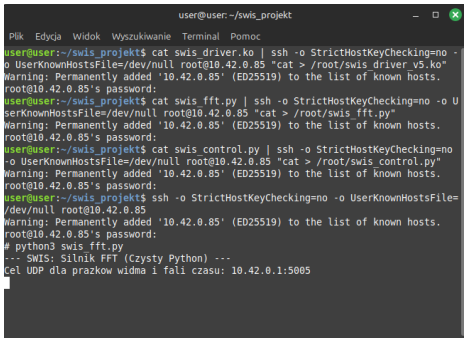
- Team collaboration allowed for the smooth merging of my embedded software with the kernel driver written by my teammate (`swis_driver.ko`).
- Proper listening of the analytical application on UDP port 5005 and lossless transmission of spectrum and time packets over the network were confirmed.
- An independent control channel was built, which applies I2C commands directly to STM32 registers (threshold change, timer pause) via the network without disrupting the readout of the main ADC data stream.



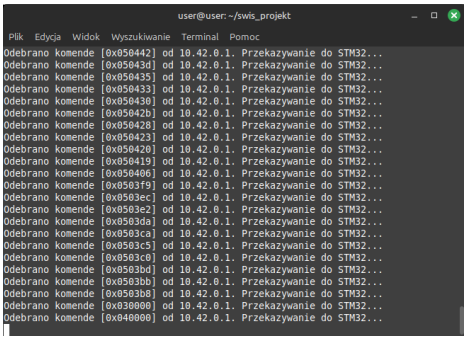
Operational acquisition system: Time domain with 0.77 V trigger threshold and FFT spectrogram



Kernel module loading and STM32 detection on the I2C bus (address 42)



Wireless deployment of DSP scripts and start of UDP transmission on port 5005



Control daemon logs of I2C commands sent to the microcontroller

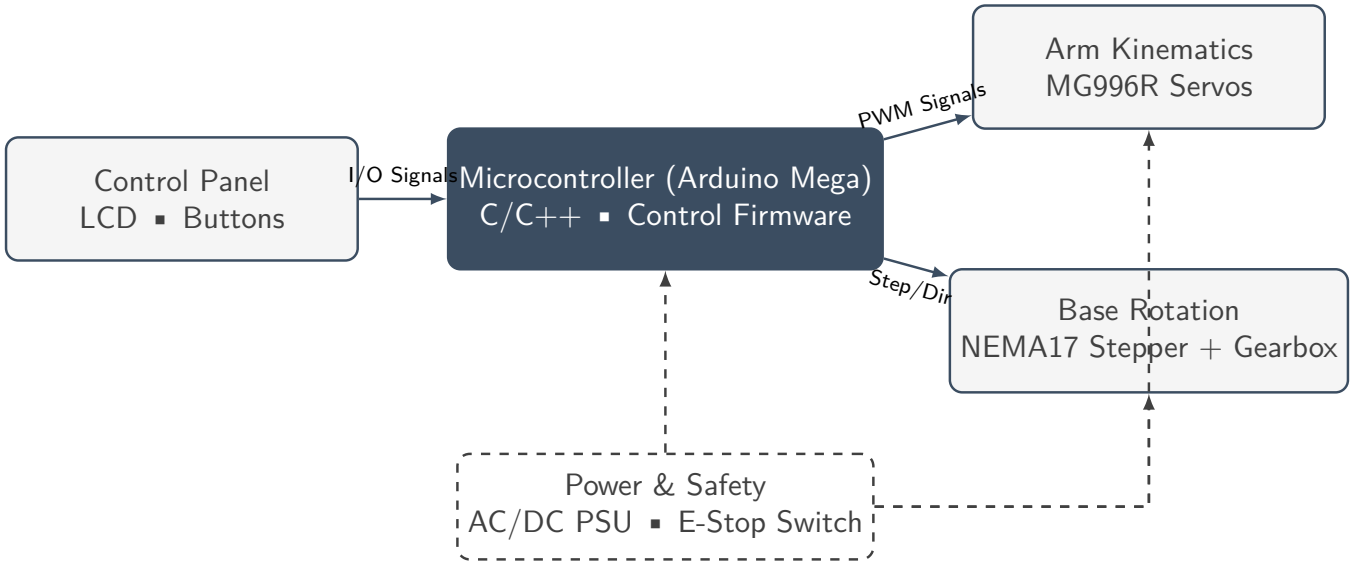
Technical High School Projects

6-Axis Robotic Arm

09.2018 - 05.2019 (Completed)

Lead Designer | Technical High School Project

The following project is a summary of the early stage of my technical education. It serves as evidence of a longstanding, deep passion for comprehensively creating mechatronic devices that require the integration of 3D mechanical design, electronics, and embedded systems programming.



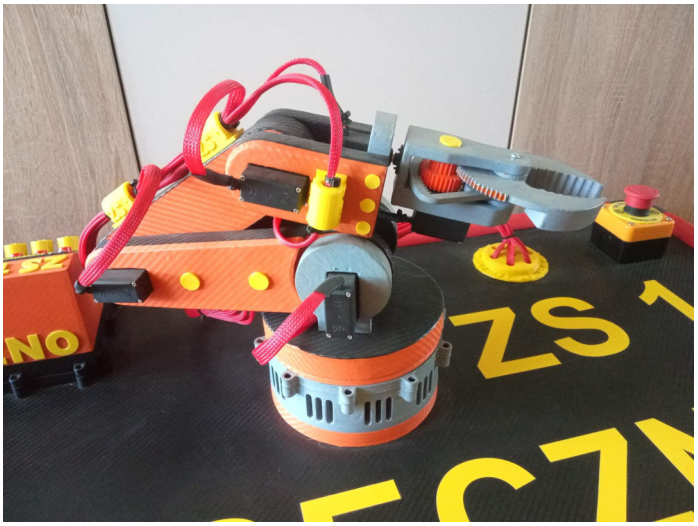
Key Solutions:

- Mechanical Design and Drive: The arm was designed from scratch in a CAD environment (DesignSpark Mechanical), and the parts were manufactured using FDM 3D printing with PLA. Kinematics and the claw are based on MG996R servos. The rotation of the entire base is achieved by a NEMA17 stepper motor with a planetary gearbox, guaranteeing smoothness and precise torque.
- Electronics and Power Supply: The heart of the system is the Arduino Mega platform. Given the early stage of learning, the power and signal distribution circuit was built on a protoboard, with all connections soldered manually. The system integrates an internal 230V AC/DC power supply with a network of external sockets.
- HMI and Safety: The user operates the robot via a dedicated panel equipped with an LCD screen, a button matrix, and a key switch. The plywood worktop (covered with carbon foil) is equipped with an active cooling system and an industrial E-Stop mushroom button. Aesthetics and durability were ensured by routing moving part cables in abrasion-resistant braided sleeves.

General view of the robot - integrated plywood base

3D printed arm structure (PLA) and claw with gears

Control panel with LCD screen and key switch



Rotary base driven by NEMA17 motor (planetary gearbox)

Cable organization in aesthetic braided sleeves and E-Stop button

View from the MG996R servos side

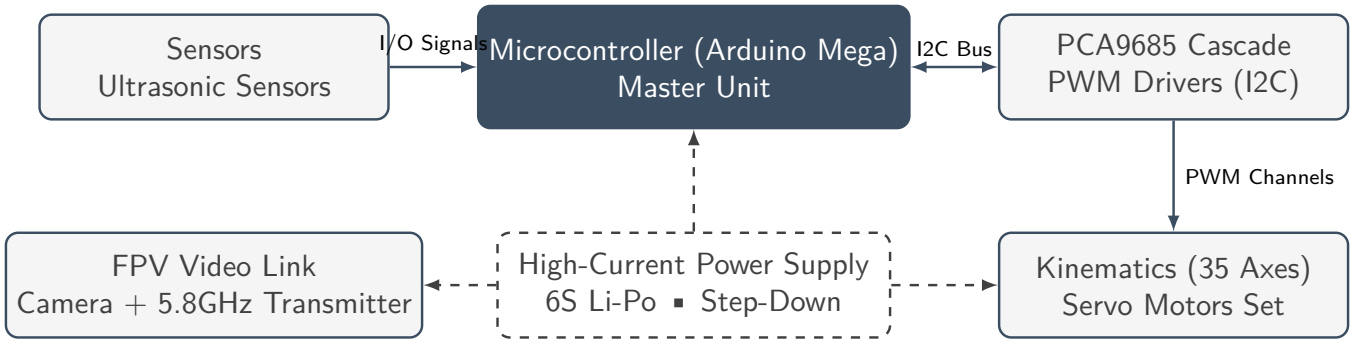
Base wall: IEC connector, active cooling, and 230V sockets

Multi-legged Walking Robot "Scorpion"

Lead Designer | Technical High School Project

2020 - 2021 (Conceptual Project)

A project from the early phase of my education, serving as evidence of a deep passion for robotics and building complex mechatronic systems. The design, inspired by scorpion anatomy, was a gigantic integration challenge requiring synchronization of 35 degrees of freedom, an FPV vision system, and high-current power distribution.



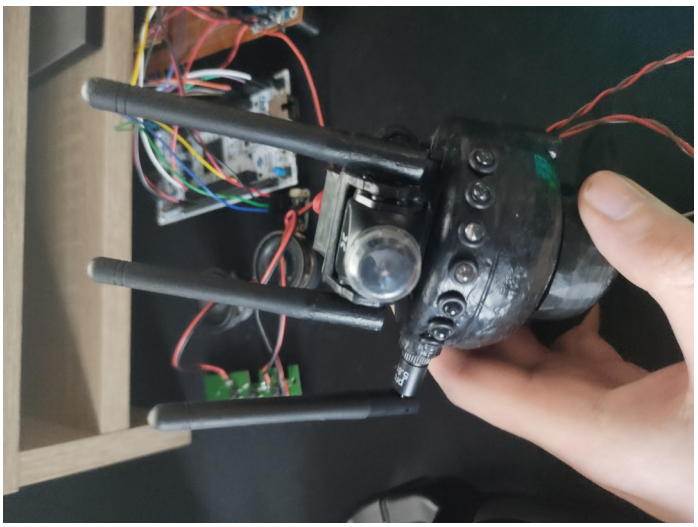
Key Design Assumptions and Architecture:

- Complex Kinematics (35 axes): Implementation of 35 independent servos responsible for the movement of 8 legs (24 servos), tail (6 servos), claws (2 servos), head (2 servos), and camera module (1 servo).
- Power Delivery: The system's power supply is based on a massive 6S Li-Po pack (over 22V), utilizing a cascade of high-current Step-Down converters to withstand enormous current inrushes during motor startup.
- Control Architecture (I2C): Utilizing the Arduino Mega platform, which communicated with a cascade of hardware PWM drivers (PCA9685 modules), minimizing processor load. An independent, analog video link (FPV transmitter/receiver) mounted in the head was also implemented.

Engineering Lesson in Humility and Future Plans:

- Mechanical Limitations: The cumulative weight of massive 3D prints and 35 servos significantly overloaded the structure. Due to insufficient torque of the utilized motors, the robot struggled to stand up efficiently. Although it could ultimately walk, its movements were very slow and lacked the desired fluidity.
- Power Integrity: Observing drastic voltage drops and interference on the I2C bus during the parallel operation of so many motors ingrained in me a rigorous approach to power budget calculations and caution in power network design.
- Project Status: Currently, the robot's development has been temporarily suspended. However, the gathered experience constitutes a powerful knowledge base – in the future, in my free time, I plan to return to this concept and build from scratch a new, significantly lighter and optimized version, devoid of the initial design flaws.

Massive, 3D printed body integrated with legs, tail, and claws



Head module with mounted FPV camera, transmitting antennas, and a servo motor

View of the upper shell part with cutouts for wiring

Robot front with specially profiled sockets for ultrasonic distance sensors